Supplementary material

Semilocal convergence beyond exact majorization for an efficient three-step method under gamma conditions

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This document brings together the algebraic derivations, rational checks, analysis of previous work, numerical experiments, and generated data for the semilocal convergence study under gamma conditions. The programs, data, and execution outputs accompany the document. Exact comparisons are distinguished from multiprecision floating-point results.

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S1 Derivation of the local error expansion

This section derives the local expansion for the three-step method. The calculation is presented by homogeneous degrees, retaining the order of the multilinear compositions. Let X, Y be Banach spaces, x_* a simple solution of $F(x) = 0$, and $F \in C^5$ in a neighborhood of x_*. We define

$$A = F'(x_*)^{-1}, \qquad H_j = \frac{1}{j!} A F^{(j)}(x_*), \quad j = 2, 3, 4, \qquad h_j = H_j(e, \dots, e), \quad e = x - x_*.$$

The estimates $O(\|e\|^k)$ are understood in the norm of X or, for linear operators, in the induced norm. Normalization at the root gives

$$AF(x_* + e) = e + h_2 + h_3 + h_4 + O(\|e\|^5). \tag{S1.1}$$

Likewise,

$$AF'(x_* + e) = I + L_1 + L_2 + L_3 + O(\|e\|^4),$$
$$L_1 v = 2H_2(e, v), \quad L_2 v = 3H_3(e, e, v), \quad L_3 v = 4H_4(e, e, e, v).$$

The Neumann series gives

$$\begin{aligned} B(e) := [AF'(x_* + e)]^{-1} = I - L_1 + (L_1^2 - L_2) \\ + (-L_3 + L_1 L_2 + L_2 L_1 - L_1^3) + O(\|e\|^4). \end{aligned} \tag{S1.2}$$

We have $F'(x_* + e)^{-1} F(x_* + u) = B(e) AF(x_* + u)$.

S1.1 First stage

Let $e_y = y - x_*$. Substituting (S1.1)–(S1.2) into the Newton step,

$$\begin{aligned} e_y = h_2 + [2h_3 - 2H_2(e, h_2)] \\ + [3h_4 - 4H_2(e, h_3) + 4H_2(e, H_2(e, h_2)) - 3H_3(e, e, h_2)] + O(\|e\|^5). \end{aligned} \tag{S1.3}$$

In particular, $e_y = O(\|e\|^2)$. Since $AF(y) = e_y + H_2(e_y, e_y) + O(\|e\|^5)$, another application of (S1.2) yields

$$\begin{aligned} B(e)AF(y) = h_2 + [2h_3 - 4H_2(e, h_2)] \\ + [3h_4 - 8H_2(e, h_3) + 12H_2(e, H_2(e, h_2)) \\ - 6H_3(e, e, h_2) + H_2(h_2, h_2)] + O(\|e\|^5). \end{aligned} \tag{S1.4}$$

S1.2 Second stage

The second point satisfies $e_z = e_y - 5B(e)AF(y)$. By the preceding identities,

$$\begin{aligned} e_z = -4h_2 + [-8h_3 + 18H_2(e, h_2)] \\ + [-12h_4 + 36H_2(e, h_3) - 56H_2(e, H_2(e, h_2)) \\ + 27H_3(e, e, h_2) - 5H_2(h_2, h_2)] + O(\|e\|^5). \end{aligned} \tag{S1.5}$$

Since $e_z = O(\|e\|^2)$, the only quadratic term contributing up to degree four in $AF(z) = e_z + H_2(e_z, e_z) + O(\|e\|^5)$ is $H_2(-4h_2, -4h_2) = 16H_2(h_2, h_2)$. Therefore,

$$\begin{aligned} AF(z) = -4h_2 + [-8h_3 + 18H_2(e, h_2)] \\ + [-12h_4 + 36H_2(e, h_3) - 56H_2(e, H_2(e, h_2)) \\ + 27H_3(e, e, h_2) + 11H_2(h_2, h_2)] + O(\|e\|^5). \end{aligned} \tag{S1.6}$$

S1.3 Third stage and fourth-degree term

Relation (S1.3) also gives $AF(y) = e_y + H_2(h_2, h_2) + O(\|e\|^5)$. Combining this with (S1.6),

$$\begin{aligned} 9AF(y) + AF(z) = 5h_2 + 10h_3 \\ + 15h_4 - 20H_2(e, H_2(e, h_2)) + 20H_2(h_2, h_2) + O(\|e\|^5). \end{aligned} \tag{S1.7}$$

Applying $B(e)/5$ and retaining the terms up to degree four,

$$\begin{aligned} \tfrac{1}{5}B(e)(9AF(y) + AF(z)) = h_2 + 2h_3 - 2H_2(e, h_2) \\ + 3h_4 - 4H_2(e, h_3) - 3H_3(e, e, h_2) + 4H_2(h_2, h_2) + O(\|e\|^5). \end{aligned} \tag{S1.8}$$

Subtracting (S1.8) from (S1.3) cancels all terms of degrees two and three and gives

$$e^+ = \mathcal{Q}_4(e) + O(\|e\|^5), \qquad \mathcal{Q}_4(e) = 4\big[H_2(e, H_2(e, H_2(e, e))) - H_2(H_2(e, e), H_2(e, e))\big]. \tag{S1.9}$$

No associativity of H_2 has been assumed. In dimension one, $H_2(p, q) = c_2pq$, and the two compositions in (S1.9) coincide; the remainder provides order at least five. In higher-dimensional spaces, this cancellation is not automatic.

For example, for $F(u, v) = (u + v^2, v + u^2)$, we have $H_2(p, q) = (p_2q_2, p_1q_1)$ and $\mathcal{Q}_4(u, v) = 4(v^3 - u^3)(u, -v)$. An iteration from $(t, 2t)$ gives

$$T(t, 2t) = (28t^4 + 224t^5, -56t^4 + 70t^5) + O(t^6). \tag{S1.10}$$

The program `code/verify_published_results.py` checks this expansion and the equivalence of the two expressions for the third stage exactly.

The condition $\mathcal{Q}_4 \not\equiv 0$ rules out uniform local order five for the entire class, but does not determine the order of each trajectory. If $e_n \neq 0$ and $\liminf \|\mathcal{Q}_4(e_n)\|/\|e_n\|^4 > 0$, the remainder bound allows $O(\|e_n\|^5)$ to be absorbed, yielding $c\|e_n\|^4 \leq \|e_{n+1}\| \leq C\|e_n\|^4$ from some index onward. The Q-order is exactly four when, in addition, the quotient has a positive finite limit. This distinction is used throughout this document.

S1.4 Exact majorization: consequence for the order and a gamma counterexample

Two assertions are distinguished. A residual bound along all indices has consequences for the complete sequence. An expansion of a single iteration in a direction proves a property of the iteration map, but does not automatically determine the order of every sequence. This distinction is retained throughout the analysis.

Consequence of exact majorization. Let $A_0 = F'(x_0)^{-1}$, $G = A_0F$, and let x_* be a simple root. If $x_n \to x_*$, differentiability and the invertibility of $G'(x_*)$ provide constants $c, C > 0$ such that

$$c\|x_n - x_*\| \leq \|G(x_n)\| \leq C\|x_n - x_*\|$$

for every sufficiently large index. If g is differentiable at t_*, $g(t_*) = 0$, $g'(t_*) < 0$, $t_n < t_*$, $t_n \uparrow t_*$, and $\delta_n = t_* - t_n$, then $g(t_n) = -g'(t_*)\delta_n + o(\delta_n)$. Therefore, $\|G(x_n)\| \leq g(t_n)$ implies $\|x_n - x_*\| \leq C_0\delta_n$ eventually.

If δ_n has Q-order at least $p > 1$, this domination gives R-order at least p, not automatically Q-order at least p. The additional bound $\delta_n \leq C_1\|x_n - x_*\|$ would suffice to obtain the corresponding Q-order inequality.

Suppose in addition that $u_n = \|x_n - x_*\| > 0$ eventually and $u_{n+1} \geq cu_n^q$ eventually, with $c > 0$ and $1 < q < p$. The scalar order gives $\delta_{n+1} \leq C\delta_n^p$ eventually. Choose N sufficiently large that both $c^{1/(q-1)}u_N$ and $C^{1/(p-1)}\delta_N$ lie in $(0, 1)$. For $a = -\log(c^{1/(q-1)}u_N) > 0$ and $b = -\log(C^{1/(p-1)}\delta_N) > 0$, iteration gives

$$u_{N+j} \geq c^{-1/(q-1)}e^{-aq^j}, \qquad \delta_{N+j} \leq C^{-1/(p-1)}e^{-bp^j}.$$

Since $q < p$, the quotient u_{N+j}/δ_{N+j} tends to infinity, contradicting exact domination. For the three-step method, the nondegenerate fourth-degree term gives $q = 4$, whereas the scalar gamma reference has $p = 5$. The trajectory-level lower bound must be checked separately; it does not follow from $\mathcal{Q}_4$ being nonzero in some direction.

A counterexample to universal validity under gamma conditions. Consider $F(u, v) = (u + v^2, v + u^2)$, the maximum norm, and $x_0 = (h, 2h)$, with $0 < h < 1/4$. We have

$$F'(x_0) = \begin{pmatrix} 1 & 4h \\ 2h & 1 \end{pmatrix}, \qquad A_0 = \frac{1}{1 - 8h^2} \begin{pmatrix} 1 & -4h \\ -2h & 1 \end{pmatrix}.$$

From $\|I - F'(x_0)\|_\infty = 4h$ and $\|F(x_0)\|_\infty = 2h + h^2$, we obtain the valid choices

$$\beta_h = \frac{2h + h^2}{1 - 4h}, \qquad \gamma_h = \frac{1}{1 - 4h}, \qquad \alpha_h = \frac{2h + h^2}{(1 - 4h)^2}. \tag{S1.11}$$

The map $F''[p, q] = (2p_2q_2, 2p_1q_1)$ has norm two and is constant. Thus $\|A_0F''\| \le 2\gamma_h$ and all its differences are zero. The gamma condition holds on the required ball, since the domain is $\mathbb{R}^2$.

The local expansion obtained in S1 gives

$$x_1 = (28h^4 + 224h^5, -56h^4 + 70h^5) + O(h^6).$$

Since $A_0 = I + O(h)$ and $F(x_1) = x_1 + O(h^8)$,

$$\|A_0F(x_1)\|_\infty = 56h^4 + O(h^5). \tag{S1.12}$$

To compute the scalar residual, we use $\widetilde{g}_\alpha(t) = \alpha - t + t^2/(1 - t)$ and denote the first complete iterate from zero by $\widehat{t}_1$. Substituting the stages of the method,

$$\widehat{s}_0 = \alpha, \qquad \widehat{v}_0 = \frac{\alpha(1 + 4\alpha)}{1 - \alpha}, \qquad \widehat{t}_1 = \frac{\alpha(1 - 2\alpha)(1 - 2\alpha^2)}{(1 - \alpha)(1 - 2\alpha - 4\alpha^2)}.$$

An exact rational simplification yields

$$\widetilde{g}_\alpha(\widehat{t}_1) = \frac{2\alpha^5(5 - 4\alpha - 8\alpha^2 + 8\alpha^3)}{(1 - \alpha)(1 - 2\alpha - 4\alpha^2)(1 - 4\alpha + 6\alpha^3 - 4\alpha^4)} = 10\alpha^5 + O(\alpha^6). \tag{S1.13}$$

With (S1.11), $\alpha_h = 2h + O(h^2)$ and $\gamma_h = 1 + O(h)$. Since $\gamma_h g_h(t_1) = \widetilde{g}_{\alpha_h}(\widehat{t}_1)$, we obtain

$$g_h(t_1) = 320h^5 + O(h^6), \qquad \frac{\|A_0F(x_1)\|_\infty}{g_h(t_1)} = \frac{7}{40h} + O(1) \longrightarrow +\infty.$$

Thus, exact majorization fails at the first iteration for every sufficiently small $h > 0$.

Rational verification for fixed data.

For $h = 1/1000$, the program `code/verify_exact_obstruction.py` computes the stages using fractions and checks

$$\alpha_h = \frac{667}{330672} < \frac{1547}{10000}, \qquad \|A_0F(x_1)\|_\infty > \frac{5}{10^{11}}, \qquad 0 < g_h(t_1) < \frac{4}{10^{13}}.$$

The approximate values, for presentation only, are

$$\|A_0F(x_1)\|_\infty \simeq 5.599035405559621\,10^{-11}, \qquad g_h(t_1) \simeq 3.367867749356755\,10^{-13}.$$

The conclusion does not depend on these decimals. The program compares integer numerators and denominators. The example lies within the interval of our criterion and refutes universal exact domination by the proposed majorant; it is not a counterexample to convergence of the method.

S1.5 Uniform local order bounds for the Lotfi maps

We examine the maps $M6$, $M9$, and $M12$ in [1], equations (5), (22), and (24), respectively. The corresponding Theorems 2.1, 3.1, and 3.2 state orders six, nine, and twelve. With $J(x) = F'(x)$, their stages can be written as

$$\begin{aligned} y &= x - J(x)^{-1}F(x), & z &= x - 2\big(J(x) + J(y)\big)^{-1}F(x), \\ B &= J(x)^{-1}J(y), & H &= \tfrac{7}{2}I - 4B + \tfrac{3}{2}B^2, \qquad K = HJ(x)^{-1}, \\ M6(x) &= z - KF(z), & M9(x) &= M6(x) - KF(M6(x)), \\ & & M12(x) &= M9(x) - KF(M9(x)). \end{aligned} \tag{S1.14}$$

The matrices H and K are kept fixed during the three corrections. In particular, the order of the matrix products in (S1.14) is not changed.

Proposition S1.1. *Let*

$$F(a,b) = (a + ab,\ b + a^2 + ab), \qquad x(t) = (t, 2t).$$

The root $x_* = 0$ is simple, all inverses in (S1.14) *exist for sufficiently small $|t|$, and*

$$\begin{aligned} M6(t,2t) &= (-\tfrac{3}{4}, \tfrac{3}{4})t^5 + O(t^6), \\ M9(t,2t) &= (\tfrac{9}{8}, -\tfrac{9}{8})t^7 + O(t^8), \\ M12(t,2t) &= (-\tfrac{27}{16}, \tfrac{27}{16})t^9 + O(t^{10}). \end{aligned} \tag{S1.15}$$

Consequently, these maps do not satisfy uniform local bounds of orders six, nine, and twelve, respectively, for all smooth systems with a simple root.

Proof. The field is polynomial, $F(0) = 0$, and $J(0) = I$. Along $x(t)$, both $J(x(t))$ and $J(x(t)) + J(y(t))$ are invertible near zero, since their values at zero are I and $2I$. Put $v = (1,2)^T$ and $q = (2,3)^T$, so that $F(tv) = tv + t^2 q$. The Jacobian and the Newton stage satisfy

$$J(x(t)) = I + tA, \qquad A = \begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix}, \qquad y(t) = t^2 q + O(t^3).$$

It follows that

$$J(y(t)) = I + t^2 C + O(t^3), \qquad C = \begin{pmatrix} 3 & 2 \\ 7 & 2 \end{pmatrix}.$$

Using $Av = 2q$ in the expansion of $2(J(x(t)) + J(y(t)))^{-1}F(x(t))$ gives

$$z(t) = \tfrac{1}{2}Cv\, t^3 + O(t^4) = \begin{pmatrix} 7/2 \\ 11/2 \end{pmatrix} t^3 + O(t^4).$$

Furthermore,

$$\begin{aligned} B &= I - tA + t^2(A^2 + C) + O(t^3), \\ H &= I + tA + t^2(\tfrac{1}{2}A^2 - C) + O(t^3), \\ I - K &= t^2 D + O(t^3), \qquad D = C - \tfrac{1}{2}A^2 = \begin{pmatrix} -1 & 1/2 \\ 1 & -1/2 \end{pmatrix}. \end{aligned}$$

Write $F(w) = w + Q(w)$, where Q is homogeneous quadratic. For a point $w(t) = c\,t^r + O(t^{r+1})$ with $r \geq 3$, the frozen correction obeys

$$w(t) - KF(w(t)) = Dc\, t^{r+2} + O(t^{r+3}),$$

since $K = I + O(t^2)$ and $Q(w(t)) = O(t^{2r})$ with $2r \geq r + 3$. Starting with $c = (7/2, 11/2)^T$ and $r = 3$, successive multiplication by D gives

$$\begin{pmatrix} -3/4 \\ 3/4 \end{pmatrix}, \qquad \begin{pmatrix} 9/8 \\ -9/8 \end{pmatrix}, \qquad \begin{pmatrix} -27/16 \\ 27/16 \end{pmatrix}.$$

This proves (S1.15). Since every leading coefficient is nonzero and $\|x(t)\|$ is proportional to $|t|$, the respective quotients by $\|x(t)\|^6$, $\|x(t)\|^9$, and $\|x(t)\|^{12}$ are unbounded as $t \to 0$. The announced uniform local estimates therefore fail for this field. □

These are statements about the iteration maps. They do not assign exact Q-orders five, seven, and nine to every complete sequence. Trajectory-level nondegeneracy must be checked separately. The same computation for the field $F_0(a, b) = (a + b^2, b + a^2)$ used in Section S1.4 instead gives

$$\begin{aligned} M6(t, 2t) &= (360, -36)t^6 + O(t^7), \\ M9(t, 2t) &= (-1512, 3024)t^8 + O(t^9), \\ M12(t, 2t) &= (12096, -24192)t^{10} + O(t^{11}). \end{aligned}$$

Thus that field, on this path, does not refute the sixth-order bound for $M6$. The two fields have distinct roles.

The program `code/verify_lotfi_orders.py` performs every series operation with `fractions.Fraction` in $\mathbb{Q}[t]/(t^{12})$. It checks all six leading coefficients, the absence of lower-degree terms, and the coefficients of $z(t)$ and $I - K$ used in the proof. Results are stored in `results/lotfi_orders.json`. The independent direct evaluations in `code/check_lotfi_numerically.py` use 200-digit arithmetic; they are consistency checks, not substitutes for the exact proof.

S2 Complete proof for the majorant sequences

This section gives the algebraic sign proof for the scalar maps. All formulas in this section correspond to the rational majorant, not to the hatted controls of the semilocal criterion.

S2.1 Majorant function and normalization

Let $\beta > 0$ and $\gamma > 0$. Consider the majorant function

$$g(u) = \beta - u + \frac{\gamma u^2}{1 - \gamma u}, \qquad 0 \leq u < \frac{1}{\gamma}.$$

We write

$$\alpha = \beta\gamma$$

and assume throughout what follows that

$$\alpha < 3 - 2\sqrt{2}.$$

We denote the small positive root of g by t_*.

We also introduce the constant

$$\rho = 1 - \frac{1}{\sqrt{2}}.$$

Lemma S2.1 (Basic signs of the majorant)**.** *We have*

$$0 < t_* < \frac{\rho}{\gamma}.$$

Moreover, for every $u \in [0, t_*)$, we have

$$g(u) > 0,$$

$$g'(u) < 0,$$

and

$$g''(u) > 0.$$

Proof. We define the normalized variable

$$x = \gamma u.$$

Then

$$G(x) = \gamma g\left(\frac{x}{\gamma}\right) = \alpha - x + \frac{x^2}{1-x}.$$

Since

$$(1-x)G(x) = \alpha - (1+\alpha)x + 2x^2,$$

the assumption

$$\alpha < 3 - 2\sqrt{2}$$

implies that the small positive root $r = \gamma t_*$ satisfies

$$0 < r < 1 - \frac{1}{\sqrt{2}} = \rho.$$

Therefore,

$$0 < t_* < \frac{\rho}{\gamma}.$$

Moreover,

$$g'(u) = -\frac{2\gamma^2 u^2 - 4\gamma u + 1}{(1-\gamma u)^2}$$

and

$$g''(u) = \frac{2\gamma}{(1-\gamma u)^3} > 0.$$

If $0 \le u < t_*$, then

$$0 \le \gamma u < \gamma t_* = r < \rho.$$

Consequently,

$$2\gamma^2 u^2 - 4\gamma u + 1 > 0.$$

It follows that

$$g'(u) < 0.$$

Finally, since t_* is the first positive root of g, we have

$$g(u) > 0, \qquad 0 \le u < t_*.$$

□

We define the auxiliary functions

$$S(u) = u - \frac{g(u)}{g'(u)},$$

$$V(u) = S(u) - 5\frac{g(S(u))}{g'(u)},$$

and

$$T(u) = S(u) - \frac{9g(S(u)) + g(V(u))}{5g'(u)}.$$

We also write

$$H(u) = g(S(u)), \qquad K(u) = g(V(u)).$$

In the normalized variable

$$x = \gamma u, \qquad r = \gamma t_*,$$

we have

$$0 < r < \rho < \frac{1}{2}.$$

Since $G(r) = 0$, we obtain

$$\alpha = r - \frac{r^2}{1-r} = \frac{r(1-2r)}{1-r}.$$

We use the same letters for the normalized functions:

$$S(x) = x - \frac{G(x)}{G'(x)},$$

$$V(x) = S(x) - 5\frac{G(S(x))}{G'(x)},$$

and

$$T(x) = S(x) - \frac{9G(S(x)) + G(V(x))}{5G'(x)}.$$

The relation between the two scales is

$$S(\gamma u) = \gamma S(u),$$

$$V(\gamma u) = \gamma V(u),$$

and

$$T(\gamma u) = \gamma T(u),$$

where the normalized functions appear on the left-hand side and the original functions on the right-hand side.

S2.2 Properties of S, H and V

Lemma S2.2 (Elementary properties of S and H)**.** *For every $u \in [0, t_*)$, we have*

$$u < S(u) < t_*.$$

Moreover,

$$H(u) > 0$$

and

$$H'(u) < 0.$$

Proof. Since

$$g(u) > 0$$

and

$$g'(u) < 0,$$

we have

$$S(u) - u = -\frac{g(u)}{g'(u)} > 0.$$

Therefore,

$$u < S(u).$$

On the other hand,

$$S(u) = u - \frac{g(u)}{g'(u)}.$$

Differentiating,

$$S'(u) = 1 - \frac{(g'(u))^2 - g(u)g''(u)}{(g'(u))^2}.$$

Hence

$$S'(u) = \frac{g(u)g''(u)}{(g'(u))^2}.$$

By the preceding lemma,

$$g(u) > 0,$$
$$g''(u) > 0,$$

and

$$(g'(u))^2 > 0.$$

Thus,

$$S'(u) > 0.$$

The function S is therefore strictly increasing on $[0, t_*]$. Moreover,

$$S(t_*) = t_* - \frac{g(t_*)}{g'(t_*)} = t_*.$$

Consequently, if $0 \le u < t_*$, then

$$S(u) < S(t_*) = t_*.$$

This proves

$$u < S(u) < t_*.$$

Now,

$$H(u) = g(S(u)).$$

Since $S(u) < t_*$ and $g > 0$ on $[0, t_*)$, we obtain

$$H(u) > 0.$$

Moreover,

$$H'(u) = g'(S(u))S'(u).$$

Since $S(u) < t_*$, we have

$$g'(S(u)) < 0.$$

Since also $S'(u) > 0$, we conclude that

$$H'(u) < 0.$$

□

Lemma S2.3 (An auxiliary polynomial for the monotonicity of V)**.** *For $0 \le x \le r$, we have*

$$R_r(x) > 0,$$

where

$$\begin{aligned} R_r(x) = {} & 32r^4x^4 - 128r^4x^3 + 216r^4x^2 - 176r^4x + 56r^4 \\ & - 128r^3x^4 + 428r^3x^3 - 612r^3x^2 + 414r^3x - 102r^3 \\ & + 204r^2x^4 - 564r^2x^3 + 708r^2x^2 - 426r^2x + 87r^2 \\ & - 152rx^4 + 314rx^3 - 318rx^2 + 169rx - 31r \\ & + 52x^4 - 82x^3 + 60x^2 - 25x + 4. \end{aligned}$$

Proof. First,

$$R_r(r) = 4(1-r)^2(2r^2-4r+1)^3 > 0,$$

because

$$2r^2 - 4r + 1 > 0.$$

Moreover,

$$R_r'(r) = (r-1)(2r^2-4r+1)^2(32r^2-64r+25).$$

Since

$$32r^2 - 64r + 25 = 16(2r^2-4r+1) + 9 > 0$$

and

$$r - 1 < 0,$$

we obtain

$$R_r'(r) < 0.$$

Differentiating three times,

$$R_r'''(x) = 12(A(r)x - B(r)),$$

where

$$A(r) = 64r^4 - 256r^3 + 408r^2 - 304r + 104$$

and

$$B(r) = 64r^4 - 214r^3 + 282r^2 - 157r + 41.$$

We have

$$A(r) = (2r^2-4r+1)(32r^2-64r+60) + 44 > 0.$$

Moreover,

$$B(r) - rA(r) = (1-r)(64r^4 - 256r^3 + 366r^2 - 220r + 41).$$

But

$$64r^4 - 256r^3 + 366r^2 - 220r + 41 = (2r^2-4r+1)(32r^2-64r+39) + 2 > 0.$$

Hence

$$B(r) - rA(r) > 0.$$

If $0 \le x \le r$, then

$$A(r)x - B(r) \le A(r)r - B(r) < 0.$$

Therefore,

$$R_r'''(x) < 0, \qquad 0 \le x \le r.$$

Thus, R_r'' is decreasing on $[0, r]$.

On the other hand,

$$R_r''(r) = 12(2r^2-4r+1)(16r^4 - 64r^3 + 91r^2 - 54r + 10).$$

The second factor satisfies

$$16r^4 - 64r^3 + 91r^2 - 54r + 10 = (2r^2-4r+1)\left(8r^2 - 16r + \frac{19}{2}\right) + \frac{1}{2} > 0.$$

It follows that

$$R_r''(r) > 0.$$

Since R_r'' is decreasing, for $0 \le x \le r$ we have

$$R_r''(x) \ge R_r''(r) > 0.$$

Hence R_r' is increasing. Since $R_r'(r) < 0$, we obtain

$$R_r'(x) \le R_r'(r) < 0, \qquad 0 \le x \le r.$$

Therefore, R_r is decreasing on $[0, r]$. Finally,

$$R_r(x) \ge R_r(r) > 0, \qquad 0 \le x \le r.$$

□

Lemma S2.4 (Monotonicity of V)**.** *The function V is strictly decreasing on $[0, r)$. In particular,*

$$V'(x) < 0, \qquad 0 \le x < r.$$

Proof. A direct calculation gives

$$V'(x) = \frac{2(r-x)(1-2r-2x+2rx)R_r(x)}{(r-1)(2x^2-4x+1)^3 D_r(x)^2},$$

where

$$D_r(x) = 2r^2x - 2r^2 - 4rx + 2r + 3x - 1.$$

That is,

$$D_r(x) = x(2r^2 - 4r + 3) - (2r^2 - 2r + 1).$$

Since

$$2r^2 - 4r + 3 > 0$$

and

$$x \le r,$$

we have

$$D_r(x) \le D_r(r).$$

Now,

$$D_r(r) = (r-1)(2r^2 - 4r + 1) < 0.$$

Hence

$$D_r(x) < 0,$$

and, in particular,

$$D_r(x)^2 > 0.$$

For $0 \le x < r$, we have

$$r - x > 0.$$

Moreover,

$$1 - 2r - 2x + 2rx \ge 1 - 4r + 2r^2 = 2r^2 - 4r + 1 > 0.$$

By the preceding lemma,

$$R_r(x) > 0.$$

Also,

$$r - 1 < 0,$$
$$2x^2 - 4x + 1 > 0,$$

and

$$D_r(x)^2 > 0.$$

Therefore, the numerator of $V'(x)$ is positive and its denominator is negative. It follows that

$$V'(x) < 0, \qquad 0 \le x < r.$$

□

Lemma S2.5 (Location of V)**.** *For every $0 \le x < r$, we have*

$$r < V(x) < 1.$$

Equivalently, for every $0 \le u < t_*$,

$$t_* < V(u) < \frac{1}{\gamma}.$$

Proof. Since

$$G(r) = 0,$$

we have

$$S(r) = r.$$

Hence

$$V(r) = S(r) - 5\frac{G(S(r))}{G'(r)} = r.$$

By the preceding lemma, V is decreasing. Therefore, if $0 \le x < r$, then

$$V(x) > V(r) = r.$$

It remains to prove that $V(x) < 1$. Since V is decreasing and $x \ge 0$, it suffices to prove

$$V(0) < 1.$$

A direct calculation gives

$$1 - V(0) = \frac{16r^4 - 12r^3 - 3r^2 + 4r - 1}{(r-1)(2r^2 - 2r + 1)}.$$

Let

$$N(r) = 16r^4 - 12r^3 - 3r^2 + 4r - 1.$$

Then

$$N'(r) = 64r^3 - 36r^2 - 6r + 4.$$

Factoring,

$$N'(r) = 2(2r-1)(16r^2 - r - 2).$$

Since

$$0 < r < \rho < \frac{1}{2},$$

we have

$$2r - 1 < 0.$$

Moreover,

$$16r^2 - r - 2 < 16\rho^2 - 2 = 22 - 16\sqrt{2} < 0.$$

Therefore,

$$N'(r) > 0.$$

Thus, N is increasing on $(0, \rho)$. Hence

$$N(r) < N(\rho).$$

Now,

$$N(\rho) = \frac{73 - 52\sqrt{2}}{2} < 0.$$

It follows that

$$N(r) < 0.$$

Since

$$r - 1 < 0$$

and

$$2r^2 - 2r + 1 > 0,$$

the denominator of $1 - V(0)$ is negative. Consequently,

$$1 - V(0) > 0.$$

This proves that

$$V(0) < 1.$$

Since $V(x) \leq V(0)$, we obtain

$$V(x) < 1.$$

Therefore,

$$r < V(x) < 1.$$

Dividing by $\gamma > 0$ gives

$$t_* < V(u) < \frac{1}{\gamma}.$$

□

S2.3 Properties of T

Lemma S2.6 (An inequality for $K = g \circ V$)**.** *For every $u \in [0, t_*)$, we have*

$$K(u) > -4H(u).$$

Consequently,

$$9H(u) + K(u) > 0.$$

Proof. Since

$$V(u) < \frac{1}{\gamma},$$

the quantity $g(V(u))$ is well defined. By the convexity of g,

$$g(V(u)) \geq g(S(u)) + g'(S(u))(V(u) - S(u)).$$

That is,

$$K(u) \geq H(u) + g'(S(u))(V(u) - S(u)).$$

But

$$V(u) - S(u) = -5\frac{g(S(u))}{g'(u)} = -5\frac{H(u)}{g'(u)}.$$

Substituting,

$$K(u) \geq H(u) - 5g'(S(u))\frac{H(u)}{g'(u)}.$$

Therefore,

$$K(u) \geq H(u)\left(1 - 5\frac{g'(S(u))}{g'(u)}\right).$$

Since

$$u < S(u) < t_*,$$

and $g'' > 0$, the function g' is strictly increasing. Thus,

$$g'(u) < g'(S(u)) < 0.$$

Dividing by $g'(u) < 0$ gives

$$0 < \frac{g'(S(u))}{g'(u)} < 1.$$

Hence

$$1 - 5\frac{g'(S(u))}{g'(u)} > -4.$$

Since $H(u) > 0$, we conclude that

$$K(u) > -4H(u).$$

Consequently,

$$9H(u) + K(u) > 9H(u) - 4H(u) = 5H(u) > 0.$$

□

Lemma S2.7 (The inequality $S < T$)**.** *For every $u \in [0, t_*)$, we have*

$$S(u) < T(u).$$

Proof. By definition,

$$T(u) - S(u) = -\frac{9g(S(u)) + g(V(u))}{5g'(u)}.$$

Using

$$H(u) = g(S(u))$$

and

$$K(u) = g(V(u)),$$

we obtain

$$T(u) - S(u) = -\frac{9H(u) + K(u)}{5g'(u)}.$$

By the preceding lemma,

$$9H(u) + K(u) > 0.$$

Moreover,

$$g'(u) < 0.$$

Therefore,

$$T(u) - S(u) > 0.$$

That is,

$$S(u) < T(u).$$

□

Lemma S2.8 (Sign of Q_r)**.** *For every $0 \le x \le r$, we have*

$$Q_r(x) > 0,$$

where

$$\begin{aligned} Q_r(x) = {} & 8r^4x^4 - 32r^4x^3 + 24r^4x^2 + 16r^4x - 16r^4 \\ & - 32r^3x^4 + 152r^3x^3 - 168r^3x^2 + 36r^3x + 12r^3 \\ & + 36r^2x^4 - 216r^2x^3 + 252r^2x^2 - 84r^2x + 3r^2 \\ & - 8rx^4 + 116rx^3 - 132rx^2 + 46rx - 4r \\ & - 2x^4 - 28x^3 + 30x^2 - 10x + 1. \end{aligned}$$

Proof. Evaluating at $x = r$ gives

$$Q_r(r) = (1-r)^2(2r^2 - 4r + 1)^3 > 0.$$

We make the change

$$y = r - x.$$

Then

$$Q_r'(r-y) = -2\widehat{H}_r(y),$$

where

$$\widehat{H}_r(y) = 4A(r)y^3 + 3B(r)y^2 + 2C(r)y + E(r),$$

with

$$A(r) = 4r^4 - 16r^3 + 18r^2 - 4r - 1,$$
$$B(r) = -16r^5 + 80r^4 - 148r^3 + 124r^2 - 54r + 14,$$
$$C(r) = 24r^6 - 144r^5 + 348r^4 - 432r^3 + 294r^2 - 108r + 15,$$

and

$$E(r) = -16r^7 + 112r^6 - 324r^5 + 500r^4 - 440r^3 + 216r^2 - 53r + 5.$$

First we prove that

$$A(r) < 0.$$

Indeed,

$$A'(r) = 4(r-1)(4r^2 - 8r + 1).$$

The function A decreases and then increases on $[0, \rho]$. Moreover,

$$A(0) = -1$$

and

$$A(\rho) = -1.$$

Consequently,

$$A(r) \le -1 < 0.$$

Now,

$$\widehat{H}_r'(y) = 12A(r)y^2 + 6B(r)y + 2C(r).$$

Since

$$A(r) < 0,$$

the function $\widehat{H}_r'$ is concave in y. Moreover,

$$\widehat{H}_r'(0) = 2C(r).$$

Factoring,

$$\widehat{H}_r'(0) = 6(2r^2 - 6r + 5)(2r^2 - 4r + 1)(2r^2 - 2r + 1) > 0.$$

Also,

$$\widehat{H}_r'(r) = 6(4r^4 - 28r^3 + 42r^2 - 22r + 5) > 0.$$

To justify the last inequality, we define

$$M(r) = 4r^4 - 28r^3 + 42r^2 - 22r + 5.$$

Then

$$M(\rho) = -7 + 6\sqrt{2} > 0.$$

Moreover,

$$M'(r) = 16r^3 - 84r^2 + 84r - 22$$

and

$$M''(r) = 48r^2 - 168r + 84 > 0$$

on $[0, \rho]$. Thus, M' is increasing on $[0, \rho]$, and therefore

$$M'(r) \le M'(\rho) = -24 + 14\sqrt{2} < 0.$$

Hence M is decreasing and, since $r < \rho$, we obtain

$$M(r) > M(\rho) > 0.$$

It follows that

$$\widehat{H}'_r(r) > 0.$$

Since $\widehat{H}'_r$ is concave and positive at the endpoints of $[0, r]$, we obtain

$$\widehat{H}'_r(y) > 0, \qquad 0 \le y \le r.$$

Hence $\widehat{H}_r$ is increasing. Moreover,

$$\widehat{H}_r(0) = E(r).$$

But

$$E(r) = -(r-1)(2r^2 - 4r + 1)^2(4r^2 - 8r + 5) > 0.$$

Therefore,

$$\widehat{H}_r(y) > 0, \qquad 0 \le y \le r.$$

Thus,

$$Q'_r(r - y) = -2\widehat{H}_r(y) < 0.$$

Hence Q_r is decreasing with respect to x. If $0 \le x < r$, then

$$Q_r(x) > Q_r(r) > 0.$$

For $x = r$, we already know that $Q_r(r) > 0$. This completes the proof. □

Lemma S2.9 (Sign of P_r)**.** *For every $0 \le x \le r$, we have*

$$P_r(x) < 0,$$

where

$$\begin{aligned} P_r(x) = {} & 8r^3x^5 - 40r^3x^4 + 56r^3x^3 - 8r^3x^2 - 32r^3x + 16r^3 \\ & + 24r^2x^6 - 168r^2x^5 + 424r^2x^4 - 424r^2x^3 + 100r^2x^2 + 72r^2x - 28r^2 \\ & - 48rx^6 + 276rx^5 - 548rx^4 + 382rx^3 - 2rx^2 - 79rx + 19r \\ & + 20x^6 - 92x^5 + 122x^4 - 6x^3 - 63x^2 + 33x - 5. \end{aligned}$$

Proof. We write

$$a = 1 - r, \qquad z = 1 - x.$$

Then

$$\frac{1}{\sqrt{2}} < a < 1, \qquad a \le z \le 1.$$

In these variables,

$$P_r(x) = \Pi(a, z),$$

where

$$\Pi(a,z) = 8a^3z^5 - 24a^3z^3 + 24a^2z^6 - 56a^2z^4 + 52a^2z^2 - 36az^5 + 62az^3 - 37az - 4z^6 + 18z^4 - 21z^2 + 9.$$

Evaluating at $z = a$, we obtain

$$\Pi(a,a) = (2a-3)(2a+3)(2a^2-1)^3 < 0.$$

Therefore, it suffices to prove that $\Pi(a,z)$ is strictly decreasing with respect to z on the interval $[a,1]$.

Differentiating,

$$\frac{\partial \Pi}{\partial z}(a,z) = 40a^3z^4 - 72a^3z^2 + 144a^2z^5 - 224a^2z^3 + 104a^2z - 180az^4 + 186az^2 - 37a - 24z^5 + 72z^3 - 42z.$$

We prove that

$$-\frac{\partial \Pi}{\partial z}(a,z) > 0.$$

Since $a \le z$, we set

$$y = \frac{a}{z}.$$

Then

$$0 < y \le 1,$$

and, since $a > 1/\sqrt{2}$, we have

$$y > \frac{1}{\sqrt{2}z}.$$

Moreover, $1/\sqrt{2} < z \le 1$. A direct calculation gives

$$-\frac{\partial \Pi}{\partial z}(a,z) = z\mathcal{L}(y,z),$$

where

$$\mathcal{L}(y,z) = -40y^3z^6 - 144y^2z^6 + 72y^3z^4 + 224y^2z^4 + 180yz^4 + 24z^4 - 104y^2z^2 - 186yz^2 - 72z^2 + 37y + 42.$$

Since $z > 0$, it suffices to prove that

$$\mathcal{L}(y,z) > 0.$$

Differentiating with respect to y,

$$\frac{\partial \mathcal{L}}{\partial y}(y,z) = -120z^6y^2 - 288z^6y + 216z^4y^2 + 448z^4y + 180z^4 - 208z^2y - 186z^2 + 37.$$

Moreover,

$$\frac{\partial^2 \mathcal{L}}{\partial y^2}(y,z) = -16z^2\left(15z^4y + 18z^4 - 27z^2y - 28z^2 + 13\right).$$

Let

$$B_z(y) = 15z^4y + 18z^4 - 27z^2y - 28z^2 + 13.$$

The coefficient of y in B_z is

$$z^2(15z^2 - 27) < 0.$$

Therefore, B_z is decreasing in y. Since

$$y \ge \frac{1}{\sqrt{2}z},$$

we have

$$B_z(y) \le B_z\left(\frac{1}{\sqrt{2}z}\right).$$

Now,

$$B_z\left(\frac{1}{\sqrt{2}z}\right) = 18z^4 + \frac{15}{\sqrt{2}}z^3 - 28z^2 - \frac{27}{\sqrt{2}}z + 13.$$

Denote

$$\Phi(z) = 18z^4 + \frac{15}{\sqrt{2}}z^3 - 28z^2 - \frac{27}{\sqrt{2}}z + 13.$$

Then

$$\Phi''(z) = 216z^2 + 45\sqrt{2}z - 56 > 0$$

for $1/\sqrt{2} \le z \le 1$. Consequently, Φ is convex. Moreover,

$$\Phi\left(\frac{1}{\sqrt{2}}\right) = -\frac{25}{4} < 0$$

and

$$\Phi(1) = 3 - 6\sqrt{2} < 0.$$

Being convex, Φ attains its maximum at the endpoints of the interval. Hence

$$\Phi(z) < 0.$$

Therefore,

$$B_z(y) < 0.$$

It follows that

$$\frac{\partial^2 \mathcal{L}}{\partial y^2}(y, z) > 0.$$

Thus, the function $\partial\mathcal{L}/\partial y$ is increasing with respect to y. Consequently,

$$\frac{\partial \mathcal{L}}{\partial y}(y, z) \ge \frac{\partial \mathcal{L}}{\partial y}\left(\frac{1}{\sqrt{2}z}, z\right).$$

A direct calculation gives

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial y}\left(\frac{1}{\sqrt{2}z}, z\right) = &-144\sqrt{2}z^5 + 120z^4 + 224\sqrt{2}z^3 \\ &- 78z^2 - 104\sqrt{2}z + 37.\end{aligned}$$

Moreover,

$$\begin{aligned}&-144\sqrt{2}z^5 + 120z^4 + 224\sqrt{2}z^3 - 78z^2 - 104\sqrt{2}z + 37 \\ &= -144\sqrt{2}\left(z - \tfrac{1}{\sqrt{2}}\right)\left(z^4 + \tfrac{\sqrt{2}}{12}z^3 - \tfrac{53}{36}z^2 - \tfrac{67\sqrt{2}}{144}z + \tfrac{37}{144}\right).\end{aligned}$$

For $1/\sqrt{2} \le z \le 1$, we have

$$\begin{aligned}&z^4 + \frac{\sqrt{2}}{12}z^3 - \frac{53}{36}z^2 - \frac{67\sqrt{2}}{144}z + \frac{37}{144} \\ &\le z^2 + \frac{\sqrt{2}}{12}z - \frac{53}{36}z^2 - \frac{67\sqrt{2}}{144}z + \frac{37}{144} \\ &= -\frac{17}{36}z^2 - \frac{55\sqrt{2}}{144}z + \frac{37}{144} \\ &\le -\frac{17}{72} - \frac{55}{144} + \frac{37}{144} < 0.\end{aligned}$$

Therefore,

$$\frac{\partial \mathcal{L}}{\partial y}\left(\frac{1}{\sqrt{2}z}, z\right) \geq 0.$$

Hence

$$\frac{\partial \mathcal{L}}{\partial y}(y, z) > 0.$$

Thus, $\mathcal{L}$ is increasing with respect to y. Consequently,

$$\mathcal{L}(y, z) \geq \mathcal{L}\left(\frac{1}{\sqrt{2}z}, z\right).$$

A direct calculation gives

$$\mathcal{L}\left(\frac{1}{\sqrt{2}z}, z\right) = -48z^4 + 80\sqrt{2}z^3 + 40z^2 - 75\sqrt{2}z - 10 + \frac{37\sqrt{2}}{2z}.$$

We define

$$\Lambda(z) = -48z^4 + 80\sqrt{2}z^3 + 40z^2 - 75\sqrt{2}z - 10 + \frac{37\sqrt{2}}{2z}.$$

Then

$$\Lambda\left(\frac{1}{\sqrt{2}}\right) = 0.$$

Moreover,

$$\begin{aligned} 2z^2\Lambda'(z) &= -384z^5 + 480\sqrt{2}z^4 + 160z^3 - 150\sqrt{2}z^2 - 37\sqrt{2} \\ &= -384\left(z - \tfrac{1}{\sqrt{2}}\right)\left(z^4 - \tfrac{3\sqrt{2}}{4}z^3 - \tfrac{7}{6}z^2 - \tfrac{37\sqrt{2}}{192}z - \tfrac{37}{192}\right). \end{aligned}$$

Since

$$z^4 - \frac{7}{6}z^2 \leq -\frac{1}{6}z^2 < 0,$$

we have

$$z^4 - \frac{3\sqrt{2}}{4}z^3 - \frac{7}{6}z^2 - \frac{37\sqrt{2}}{192}z - \frac{37}{192} < 0.$$

Therefore,

$$\Lambda'(z) \geq 0$$

for $1/\sqrt{2} \leq z \leq 1$. Since $z > 1/\sqrt{2}$, we conclude that

$$\Lambda(z) > \Lambda\left(\frac{1}{\sqrt{2}}\right) = 0.$$

Thus,

$$\mathcal{L}(y, z) > 0.$$

Consequently,

$$-\frac{\partial \Pi}{\partial z}(a, z) > 0.$$

Hence

$$\frac{\partial \Pi}{\partial z}(a, z) < 0.$$

Therefore, $\Pi(a, z)$ is strictly decreasing with respect to z. Since $z \geq a$, we obtain

$$\Pi(a, z) \leq \Pi(a, a) < 0.$$

Thus,

$$P_r(x) < 0.$$

□

Lemma S2.10 (The inequality $T(x) < r$)**.** *For every $0 \le x < r$, we have*

$$T(x) < r.$$

Equivalently, for every $0 \le u < t_*$,

$$T(u) < t_*.$$

Proof. A direct algebraic calculation gives

$$T(x) - r = \frac{2(r-x)^5 P_r(x)}{(r-1)(2x^2-4x+1)^3 D_r(x) Q_r(x)},$$

where

$$D_r(x) = 2r^2x - 2r^2 - 4rx + 2r + 3x - 1.$$

We have already proved that

$$D_r(x) < 0,$$

$$Q_r(x) > 0,$$

and

$$P_r(x) < 0.$$

Moreover, for $0 \le x < r$,

$$(r-x)^5 > 0.$$

Therefore, the numerator of $T(x) - r$ is negative.

On the other hand,

$$r - 1 < 0,$$

$$2x^2 - 4x + 1 > 0,$$

$$D_r(x) < 0,$$

and

$$Q_r(x) > 0.$$

Hence the denominator is positive. Consequently,

$$T(x) - r < 0.$$

That is,

$$T(x) < r.$$

Dividing by $\gamma > 0$ gives

$$T(u) < t_*.$$

□

S2.4 Functional proposition and scalar sequence

Proposition S2.11 (Functional properties)**.** *For every $u \in [0, t_*)$, the following properties hold:*

$$u < S(u) < t_*,$$

$$H(u) > 0, \qquad H'(u) < 0,$$

$$t_* < V(u) < \frac{1}{\gamma},$$

$$K(u) > -4H(u),$$

$$9H(u) + K(u) > 0,$$

$$S(u) < T(u) < t_*,$$

and

$$V'(u) < 0.$$

In particular, V is strictly decreasing on $[0, t_*)$.

Proof. The first two properties follow from Lemma S2.2. The location of V follows from Lemma S2.5. The inequalities

$$K(u) > -4H(u)$$

and

$$9H(u) + K(u) > 0$$

follow from Lemma S2.6. The inequality

$$S(u) < T(u)$$

follows from Lemma S2.7, while

$$T(u) < t_*$$

follows from Lemma S2.10. Finally, the strict monotonicity of V follows from Lemma S2.4, transferring the inequality from the normalized variable x to the original variable u. □

We define

$$t_0 = 0,$$

$$s_n = S(t_n),$$

$$v_n = V(t_n),$$

and

$$t_{n+1} = T(t_n).$$

Equivalently,

$$s_n = t_n - \frac{g(t_n)}{g'(t_n)},$$

$$v_n = s_n - 5\frac{g(s_n)}{g'(t_n)},$$

and

$$t_{n+1} = s_n - \frac{9g(s_n) + g(v_n)}{5g'(t_n)}.$$

Theorem S2.12 (Interlacing of the sequences)**.** *The preceding sequence is well defined and, for every $n \geq 0$, satisfies*

$$0 \leq t_n \leq s_n \leq t_{n+1} \leq s_{n+1} < t_* < v_{n+1} \leq v_n.$$

Moreover,

$$t_* < v_n < \frac{1}{\gamma}, \qquad n \geq 0.$$

Proof. We proceed by induction.

Since

$$t_0 = 0 < t_*,$$

we can apply the preceding proposition with $u = t_0$. We obtain

$$t_0 < S(t_0) < T(t_0) < t_*.$$

Since

$$s_0 = S(t_0)$$

and

$$t_1 = T(t_0),$$

we have

$$0 \le t_0 \le s_0 \le t_1 < t_*.$$

Applying the proposition again, now with $u = t_1$, gives

$$t_1 \le s_1 < t_*.$$

We also obtain

$$t_* < v_1 < \frac{1}{\gamma}.$$

On the other hand, the proposition applied at $u = t_0$ gives

$$t_* < v_0 < \frac{1}{\gamma}.$$

Since V is decreasing and

$$t_0 \le t_1,$$

we deduce

$$v_1 = V(t_1) \le V(t_0) = v_0.$$

Therefore,

$$0 \le t_0 \le s_0 \le t_1 \le s_1 < t_* < v_1 \le v_0.$$

This proves the base case.

Now suppose that, for some $n \ge 0$, we have

$$0 \le t_n < t_*.$$

Applying the proposition with $u = t_n$, we obtain

$$t_n < S(t_n) < T(t_n) < t_*.$$

Since

$$s_n = S(t_n)$$

and

$$t_{n+1} = T(t_n),$$

we obtain

$$0 \le t_n \le s_n \le t_{n+1} < t_*.$$

In particular,

$$0 \le t_{n+1} < t_*.$$

Applying the proposition again, now with $u = t_{n+1}$, gives

$$t_{n+1} \le S(t_{n+1}) < t_*.$$

Since

$$s_{n+1} = S(t_{n+1}),$$

we have

$$t_{n+1} \le s_{n+1} < t_*.$$

Moreover, by the property of V,

$$t_* < V(t_{n+1}) < \frac{1}{\gamma}.$$

That is,

$$t_* < v_{n+1} < \frac{1}{\gamma}.$$

Finally, since V is decreasing and

$$t_n \le t_{n+1},$$

it follows that

$$v_{n+1} = V(t_{n+1}) \le V(t_n) = v_n.$$

Collecting the preceding inequalities, we conclude that

$$0 \le t_n \le s_n \le t_{n+1} \le s_{n+1} < t_* < v_{n+1} \le v_n.$$

Moreover, at each step we have

$$v_n < \frac{1}{\gamma},$$

so all terms $g(v_n)$ appearing in the definition of t_{n+1} are well defined. This completes the induction. □

S2.5 Convergence and asymptotic constants

For the following references, we write the chain already proved as

$$0 \le t_n \le s_n \le t_{n+1} \le s_{n+1} < t_* < v_{n+1} \le v_n < 1/\gamma. \tag{S2.1}$$

Theorem S2.13 (Convergence and asymptotic constants)**.** *The sequences in Theorem S2.12 satisfy $t_n \uparrow t_*$, $s_n \uparrow t_*$, and $v_n \downarrow t_*$. Let $r = \gamma t_*$ and let $d = 2r^2 - 4r + 1 > 0$. Then*

$$\begin{aligned} &\lim_{n\to\infty} \frac{t_* - s_n}{(t_* - t_n)^2} = \frac{\gamma}{(1-r)d}, \qquad \lim_{n\to\infty} \frac{v_n - t_*}{(t_* - t_n)^2} = \frac{4\gamma}{(1-r)d}, \\ &\lim_{n\to\infty} \frac{t_* - t_{n+1}}{(t_* - t_n)^5} = \frac{2\gamma^4(1+2r)(5-2r)}{(1-r)^4 d^4}. \end{aligned} \tag{S2.2}$$

In particular, there exist $C_s, C_v, C_5 > 0$ and $n_0 \ge 0$ such that, for $n \ge n_0$,

$$\begin{aligned} &0 \le t_* - s_n \le C_s(t_* - t_n)^2, \qquad 0 \le v_n - t_* \le C_v(t_* - t_n)^2, \\ &0 \le t_* - t_{n+1} \le C_5(t_* - t_n)^5. \end{aligned} \tag{S2.3}$$

Proof. Convergence of the three sequences. The chain (S2.1) shows that $\{t_n\}$ and $\{s_n\}$ are increasing and bounded above by t_*, while $\{v_n\}$ is decreasing and bounded below by t_*. Let $\ell = \lim_{n\to\infty} t_n \le t_*$. The maps S, V, T extend continuously to t_* because $g'(t_*) \ne 0$, and satisfy $S(t_*) = V(t_*) = T(t_*) = t_*$. Taking the limit in $t_{n+1} = T(t_n)$ gives $\ell = T(\ell)$. If $\ell < t_*$, Proposition S2.11 would give $T(\ell) > \ell$, a contradiction. Therefore, $\ell = t_*$. Continuity now gives

$$s_n = S(t_n) \longrightarrow S(t_*) = t_*, \qquad v_n = V(t_n) \longrightarrow V(t_*) = t_*.$$

Asymptotic constants. To distinguish the scales explicitly in this calculation, we write $\widetilde{g} = G$ and

$$\widehat{S}(x) = \gamma S(x/\gamma), \quad \widehat{V}(x) = \gamma V(x/\gamma), \quad \widehat{T}(x) = \gamma T(x/\gamma),$$

where the right-hand sides contain the original functions of g. The hatted functions are the normalized maps used in the preceding factorizations. With $x = \gamma t$ and $r = \gamma t_*$, recall that

$$\widetilde{g}'(r) = -\frac{d}{(1-r)^2}, \qquad \widetilde{g}''(r) = \frac{2}{(1-r)^3}, \qquad d = 2r^2 - 4r + 1 > 0.$$

Set $\delta = r - x$ and $a = 1/((1-r)d)$. The expansion of the Newton step at the simple zero r gives

$$r - \widehat{S}(x) = -\frac{\widetilde{g}''(r)}{2\widetilde{g}'(r)}\delta^2 + O(\delta^3) = a\delta^2 + O(\delta^3).$$

In particular, $\widehat{S}(x) - r = O(\delta^2)$. Expanding $\widetilde{g}$ at r and using $\widetilde{g}'(x) = \widetilde{g}'(r) + O(\delta)$ yields

$$\frac{\widetilde{g}(\widehat{S}(x))}{\widetilde{g}'(x)} = \widehat{S}(x) - r + O(\delta^3) = -a\delta^2 + O(\delta^3).$$

Substituting into the second stage,

$$\widehat{V}(x) - r = \widehat{S}(x) - r - 5\frac{\widetilde{g}(\widehat{S}(x))}{\widetilde{g}'(x)} = 4a\delta^2 + O(\delta^3).$$

For the third stage, the expansion at $x = r$ of the exact factorization of $\widehat{T}(x) - r$ used in the preceding proof gives

$$r - \widehat{T}(x) = \frac{2(1+2r)(5-2r)}{(1-r)^4 d^4}\delta^5 + O(\delta^6).$$

The identities yielding this constant are

$$D_r(r) = (r-1)d, \qquad Q_r(r) = (1-r)^2 d^3, \qquad P_r(r) = (2r-5)(2r+1)d^3.$$

They are checked by substituting $x = r$ into the preceding polynomials. The program `code/symbolic_majorant_verification.py` verifies these three identities and the complete factorization of $T(x) - r$.

To return to the original sequences, we take $x = \gamma t_n$, divide each expansion by γ, and substitute $\delta = \gamma(t_* - t_n)$. The constants of the quadratic terms are multiplied by γ, and that of the fifth-degree term by γ^4. This yields the three limits in (S2.2). All are positive and finite, since $0 < r < 1 - 1/\sqrt{2}$. Finally, each quotient is bounded by a positive constant from some index onward; taking a common index gives the three estimates in (S2.3). □

S2.6 Summability of the scalar controls

To transfer the scalar control to the problem in the Banach space, we will need the lengths of the three stages and the residual combination appearing in the last step. We define

$$\begin{aligned} &a_n = s_n - t_n, \qquad b_n = v_n - s_n, \qquad c_n = t_{n+1} - s_n, \\ &D_n = 9g(s_n) + g(v_n). \end{aligned}$$

Corollary S2.14 (Summability of the scalar controls)**.** *For every $p > 0$,*

$$\sum_{n=0}^{\infty} (t_* - t_n)^p < \infty.$$

Moreover,

$$\begin{aligned} &\sum_{n=0}^{\infty} a_n \le t_*, \qquad \sum_{n=0}^{\infty} c_n \le t_*, \qquad \sum_{n=0}^{\infty} b_n < \infty, \qquad \sum_{n=0}^{\infty} b_n^2 < \infty, \\ &\sum_{n=0}^{\infty} g(t_n) \le t_*, \qquad \sum_{n=0}^{\infty} D_n \le 5t_*. \end{aligned} \tag{S2.4}$$

Proof. Let $e_n = t_* - t_n$. By (S2.3), there exists n_1 such that $e_{n+1} \le C_5 e_n^5$ and $C_5 e_n^4 \le 1/2$ for $n \ge n_1$. Hence $e_{n+1} \le e_n/2$, and geometric comparison gives the convergence of $\sum e_n^p$ for every $p > 0$.

Interlacing gives $0 \le a_n, c_n \le t_{n+1} - t_n$; on summing, the first two bounds in (S2.4) are telescopic. Moreover,

$$b_n = (v_n - t_*) + (t_* - s_n) = O(e_n^2),$$

and the summability of e_n^2 and e_n^4 gives the assertions about b_n. Finally,

$$g(t_n) = (-g'(t_n))a_n \le a_n, \qquad D_n = 5(-g'(t_n))c_n \le 5c_n,$$

because $0 < -g'(t_n) \le 1$. Summing gives the two remaining bounds. □

S3 Radius controls, first cycle, and quantitative comparison

This section develops the rational verification of the criterion $0 < \alpha = \beta\gamma \le 1547/10000$ and the quantitative comparisons. The rational recurrences τ_n, ε_n are distinguished from the sequences $\widehat{\tau}_n, \widehat{\varepsilon}_n$, which use a specific estimate for the first cycle. Both retain the same scalar majorant g and the same method; the hats do not denote a change of algorithm.

S3.1 Assumptions and estimates in the normalized scale

Let $G(x) = F'(x_0)^{-1}F(x)$, $\rho = 1 - 1/\sqrt{2}$, $R_\gamma = \rho/\gamma$, and $g(t) = \beta - t + \gamma t^2/(1-\gamma t)$. We assume $B[x_0, R_\gamma] \subset \Omega$, $\|G(x_0)\| \le \beta$, $\|G''(x_0)\| \le 2\gamma$, and, for $x, z \in B[x_0, R_\gamma]$ and $u, v \in [0, R_\gamma]$ with $\|x - z\| \le |u - v|$,

$$\|G''(x) - G''(z)\| \le |g''(u) - g''(v)|.$$

Set $\mathcal{F}(\xi) = \gamma G(x_0 + \xi/\gamma)$. Then $\mathcal{F}'(0) = I$, $\|\mathcal{F}(0)\| \le \alpha = \beta\gamma$, and $\|\mathcal{F}''(0)\| \le 2$. The difference condition above implies

$$\|\mathcal{F}''(\xi) - \mathcal{F}''(\zeta)\| \le 6\|\xi - \zeta\|, \qquad \xi, \zeta \in B[0, \rho]. \tag{S3.1}$$

Indeed, subdividing the segment between the points into N parts and applying the gamma condition with parameters d/N and 0 gives $N(M(d/N) - M(0)) \to 6d$, where $d = \|\xi - \zeta\|$ and $M(t) = 2/(1-t)^3$. Choose N sufficiently large that $d/N \le \rho$, so both parameters belong to $[0, \rho]$. The converse uses $M'(t) \ge 6$. Only $F \in C^2$ is assumed, not the existence of F'''.

The rational functions used in the cycles are

$$k(t) = \frac{t^2}{1-t}, \qquad m(t) = 2 - \frac{1}{(1-t)^2}, \qquad M(t) = \frac{2}{(1-t)^3},$$
$$L(t,d) = \frac{d(2(1-t)-d)}{(1-t)^2(1-t-d)^2}, \quad E_2(t,d) = \frac{d^2}{(1-t)^2(1-t-d)}, \quad E_3(t,d) = \frac{d^3}{(1-t)^3(1-t-d)}. \tag{S3.2}$$

For $0 \le t < \rho$, $m(t) > 0$. If $\|\xi\| \le t$, then $\|\mathcal{F}'(\xi)^{-1}\| \le 1/m(t)$. The functions L, E_2, E_3 control, respectively, the variation of the derivative, the first-order remainder, and the remainder after subtracting the quadratic term. Their integral representations are

$$L(t,d) = \int_0^d M(t+s)\,ds, \quad E_2(t,d) = \int_0^d (d-s)M(t+s)\,ds,$$
$$E_3(t,d) = \int_0^d (d-s)(M(t+s) - M(t))\,ds.$$

They apply when $t + d \le \rho$ and give

$$\begin{aligned} \|\mathcal{F}'(\xi+h) - \mathcal{F}'(\xi)\| &\le L(t,d), \\ \|\mathcal{F}(\xi+h) - \mathcal{F}(\xi) - \mathcal{F}'(\xi)h\| &\le E_2(t,d), \\ \|\mathcal{F}(\xi+h) - \mathcal{F}(\xi) - \mathcal{F}'(\xi)h - \tfrac{1}{2}\mathcal{F}''(\xi)[h,h]\| &\le E_3(t,d), \end{aligned} \tag{S3.3}$$

whenever $\|\xi\| \le t$ and $\|h\| \le d$. The monotonicity and convexity of M prove that L, E_2, E_3 are nondecreasing in each argument. When $t + d \le \sigma < \rho$, we have $L(t,d) \le M(\sigma)d$ and $E_2(t,d) \le M(\sigma)d^2/2$.

S3.2 Residual identities and rational recurrences

For a cycle, we write $J = \mathcal{F}'(\xi)$, $p = y - \xi$, $q = z - y$, $r = \xi^+ - y$, $w = r - q/5$, $D = \mathcal{F}'(y) - J$, and

$$\mathcal{B}(d) = \mathcal{F}(y+d) - \mathcal{F}(y) - \mathcal{F}'(y)d, \qquad \mathcal{E}(d) = \mathcal{B}(d) - \tfrac{1}{2}\mathcal{F}''(y)[d,d].$$

The relations $Jq = -5\mathcal{F}(y)$ and $Jr = -(9\mathcal{F}(y) + \mathcal{F}(z))/5$ give the identities

$$\begin{aligned} w &= -\tfrac{1}{5}J^{-1}(Dq + \mathcal{B}(q)), \\ \mathcal{F}(\xi^+) &= Dw + \mathcal{B}(r) - \tfrac{1}{5}\mathcal{B}(q) \\ &= Dw + \tfrac{1}{2}\mathcal{F}''(y)[w,w] + \tfrac{1}{5}\mathcal{F}''(y)[w,q] - \tfrac{2}{25}\mathcal{F}''(y)[q,q] + \mathcal{E}(r) - \tfrac{1}{5}\mathcal{E}(q). \end{aligned} \tag{S3.4}$$

The last equality uses $r = w + q/5$ and bilinear symmetry, without reassociating compositions of $\mathcal{F}''$. First, y, z are placed in the domain; the first identity controls ξ^+ before its residual is evaluated.

Given controls (τ_n, ε_n), the rational recurrences are

$$\begin{aligned} m_n &= m(\tau_n), & A_n &= \varepsilon_n/m_n, & S_n &= \tau_n + A_n, \\ Y_n &= E_2(\tau_n, A_n), & B_n &= 5Y_n/m_n, & V_n &= S_n + B_n, \\ \ell_n &= L(\tau_n, A_n), & W_n &= \frac{\ell_n B_n + E_2(S_n, B_n)}{5m_n}, & C_n &= B_n/5 + W_n, \\ \tau_{n+1} &= S_n + C_n. \end{aligned} \tag{S3.5}$$

We define

$$\begin{aligned} P_n^{(2)} &= \ell_n W_n + E_2(S_n, C_n) + \tfrac{1}{5}E_2(S_n, B_n), \\ P_n^{(3)} &= \ell_n W_n + M(S_n)\left(\tfrac{1}{2}W_n^2 + \tfrac{1}{5}W_n B_n + \tfrac{2}{25}B_n^2\right) + E_3(S_n, C_n) + \tfrac{1}{5}E_3(S_n, B_n), \\ \varepsilon_{n+1} &= \min\{P_n^{(2)}, P_n^{(3)}\}. \end{aligned} \tag{S3.6}$$

A cycle is admissible if its quantities are constructed in the indicated order, $\varepsilon_n \geq 0$, and $\tau_n, S_n, V_n, \tau_{n+1} < \rho$. For an admissible cycle with $\|\xi_n\| \leq \tau_n$ and $\|\mathcal{F}(\xi_n)\| \leq \varepsilon_n$, the inverse bound, (S3.3), and (S3.4) give

$$\begin{aligned} &\|\bar{p}_n\| \leq A_n, \quad &&\|\upsilon_n\| \leq S_n, \quad &&\|\mathcal{F}(\upsilon_n)\| \leq Y_n, \\ &\|\bar{q}_n\| \leq B_n, \quad &&\|\zeta_n\| \leq V_n, \quad &&\|\bar{r}_n - \bar{q}_n/5\| \leq W_n, \\ &\|\bar{r}_n\| \leq C_n, \quad &&\|\xi_{n+1}\| \leq \tau_{n+1}, \quad &&\|\mathcal{F}(\xi_{n+1})\| \leq \varepsilon_{n+1}. \end{aligned} \tag{S3.7}$$

Monotonicity of the controls follows successively from that of $1/m, L, E_2, E_3$ and the nonnegativity of all products; the minimum of two nondecreasing quantities is also nondecreasing. Thus, the upper data of an admissible cycle dominate those of any cycle started below them.

S3.3 Specific estimate for the first cycle

We fix

$$\alpha_c = \frac{1547}{10000}, \qquad \sigma_{\mathrm{all}} = \frac{3661}{12500}, \qquad \sigma_{\mathrm{tail}} = \frac{6}{25}.$$

For the initial cycle, we use the sharper consequence of (S3.1), namely $\|\mathcal{F}''(\xi)\| \leq 2 + 6\|\xi\|$. By integration, the three remainder estimates in (S3.3) also hold, for $t + d \leq \rho$, with the controls

$$\begin{aligned} M_*(t) &= 2 + 6t, & L_*(t,d) &= (2+6t)d + 3d^2, \\ E_{2,*}(t,d) &= (1+3t)d^2 + d^3, & E_{3,*}(d) &= d^3. \end{aligned} \tag{S3.8}$$

Indeed, we integrate $2 + 6t + 6\theta d$ and $(1-\theta)(2+6t+6\theta d)$, respectively; for the second-order remainder we integrate $6\theta(1-\theta)d^3$. No F''' is required.

We use hats to distinguish the sequences that employ these bounds only at $n = 0$. We define

$$\begin{aligned}
&\widehat{\tau}_0 = 0, && \widehat{\varepsilon}_0 = \widehat{A}_0 = \widehat{S}_0 = \alpha, && \widehat{m}_0 = 1,\\
&\widehat{Y}_0 = \alpha^2 + \alpha^3, && \widehat{B}_0 = 5\widehat{Y}_0, && \widehat{V}_0 = \alpha + \widehat{B}_0,\\
&\widehat{\ell}_0 = 2\alpha + 3\alpha^2, && \widehat{W}_0 = \frac{\widehat{\ell}_0\widehat{B}_0 + E_{2,*}(\alpha, \widehat{B}_0)}{5}, && \widehat{C}_0 = \widehat{Y}_0 + \widehat{W}_0,\\
&\widehat{\tau}_1 = \alpha + \widehat{C}_0.
\end{aligned} \tag{S3.9}$$

The next residual is controlled by

$$\widehat{\varepsilon}_1 = \widehat{\ell}_0\widehat{W}_0 + (2 + 6\alpha)\left(\frac{\widehat{W}_0^2}{2} + \frac{\widehat{W}_0\widehat{B}_0}{5} + \frac{2\widehat{B}_0^2}{25}\right) + \widehat{C}_0^3 + \frac{\widehat{B}_0^3}{5}. \tag{S3.10}$$

For $n \geq 1$, we use exactly (S3.5)–(S3.6), placing a hat on every quantity in the cycle. The functions m, M, L, E_2, E_3 and the iteration of F do not change.

Lemma S3.1 (Control of the first cycle)**.** *If F satisfies the gamma condition and $0 < \alpha \leq \alpha_c$, the initial stages are well defined and satisfy* (S3.7) *with the controls in* (S3.9)–(S3.10)*. In particular,*

$$\widehat{V}_0 \leq 0.292871921615 < \sigma_{\text{all}} < \rho, \qquad \widehat{\tau}_1 < \frac{1}{5}, \qquad \widehat{\varepsilon}_1 < \frac{1}{75}. \tag{S3.11}$$

Proof. Since $\mathcal{F}'(0) = I$, $\|\upsilon_0\| \leq \alpha$, and the Newton cancellation gives

$$\|\mathcal{F}(\upsilon_0)\| \leq \alpha^2 \int_0^1 (1-\theta)(2 + 6\theta\alpha)\, d\theta = \alpha^2 + \alpha^3 = \widehat{Y}_0.$$

Then $\|\bar{q}_0\| \leq \widehat{B}_0$ and $\|\zeta_0\| \leq \widehat{V}_0 = \alpha + 5\alpha^2 + 5\alpha^3$. This polynomial is increasing, and its value at α_c is 0.292871921615. The inequality $\sigma_{\text{all}} < \rho$ is checked exactly through $2(1 - \sigma_{\text{all}})^2 - 1 = 2921/78125000 > 0$. Thus, $\mathcal{F}(\zeta_0)$ can be evaluated within the domain.

The first identity in (S3.4), together with (S3.8), gives $\|\bar{r}_0 - \bar{q}_0/5\| \leq \widehat{W}_0$ and $\|\bar{r}_0\| \leq \widehat{C}_0$. Thus, $\|\xi_1\| \leq \widehat{\tau}_1$ before its residual is evaluated. All quantities in (S3.9)–(S3.10) are polynomials in α with nonnegative coefficients. Their rational evaluation at α_c gives

$$\widehat{\tau}_1 < 0.198986457 < 1/5, \qquad \widehat{\varepsilon}_1 < 0.013187894 < 1/75.$$

The first bound also places the segment from υ_0 to ξ_1 inside the ball. Applying (S3.4), with the remainders bounded by $\widehat{C}_0^3$ and $\widehat{B}_0^3$, then gives $\|\mathcal{F}(\xi_1)\| \leq \widehat{\varepsilon}_1$. The fractions are recorded in the results file described below. □

We also define

$$\begin{aligned}
\widehat{\Phi}_{\text{ef}}(\alpha) &= \sup_{n \geq 0} \max\{\widehat{V}_n, \widehat{\tau}_{n+1}\},\\
\widehat{\Delta}_n &= \frac{[\widehat{\varepsilon}_n - \gamma g(t_n)]_+}{\gamma}, \qquad \widehat{E}_n = \frac{[\widehat{A}_n + \widehat{C}_n - \gamma(t_{n+1} - t_n)]_+}{\gamma}.
\end{aligned} \tag{S3.12}$$

The same admissibility convention is adopted for $\widehat{\Phi}_{\text{ef}}$. For $0 < \alpha \leq \alpha_c$, (S3.7) holds at $n = 0$ by Lemma S3.1 and for $n \geq 1$ by the standard cycle estimates above. Moreover, $\widehat{\tau}_{n+1} - \widehat{\tau}_n = \widehat{A}_n + \widehat{C}_n$ and $\widehat{\varepsilon}_n \leq \widehat{A}_n$ also hold at $n = 0$. Under the corresponding hypotheses expressed with the hatted controls, in particular $\widehat{\Phi}_{\text{ef}}(\alpha) \leq \sigma < \rho$, the same arguments apply. This global bound is verified in Section S3.5. It gives telescopic summability of $\widehat{A}_n + \widehat{C}_n$ and of the corresponding nonnegative defects. The one-step bounds then make $\{x_n\}$ a Cauchy sequence; continuity gives a zero of F, and the standard averaged-derivative argument with $k'(t) < 1$ gives uniqueness in the gamma ball. The same telescoping argument yields the a priori error bounds, while the residual-distance estimate gives the residual error bound. These are the properties used in the finite tail verification below, which starts at an index $N \geq 1$.

S3.4 Finite verification and control of all remaining indices

The tail control is based on the uniform constants $m = m(\sigma)$, $M = M(\sigma)$ and the polynomials

$$\begin{aligned} a(e) &= e/m, & b(e) &= \frac{5M}{2m}a(e)^2, \\ w(e) &= \frac{Ma(e)b(e) + Mb(e)^2/2}{5m}, & c(e) &= b(e)/5 + w(e), \\ P(e) &= Ma(e)w(e) + \frac{M}{2}(c(e)^2 + b(e)^2/5). \end{aligned} \tag{S3.13}$$

We have

$$P(e) = \frac{5M^3}{4m^6}e^4 + \frac{7M^4}{8m^8}e^5 + \frac{7M^5}{16m^{10}}e^6 + \frac{5M^6}{16m^{12}}e^7 + \frac{25M^7}{128m^{14}}e^8. \tag{S3.14}$$

The program `code/verify_algebra.py` checks this identity symbolically.

If $\widehat{\tau}_N \le T$ and $\widehat{\varepsilon}_N \le e$, it suffices to find $q \in (0,1)$ and $\sigma < \rho$ such that

$$P(e) \le qe, \qquad T + \max\{(a(e) + c(e))/(1-q), a(e) + b(e)\} < \sigma. \tag{S3.15}$$

To explain why all indices are controlled, set $D = a(e) + c(e)$, $H = a(e) + b(e)$. The nonnegative coefficients give $a(q^j e) + c(q^j e) \le q^j D$, $a(q^j e) + b(q^j e) \le q^j H$, and $P(q^j e) \le q^j P(e)$. Assuming inductively that

$$\widehat{\tau}_{N+j} \le T + D\frac{1-q^j}{1-q}, \qquad \widehat{\varepsilon}_{N+j} \le q^j e,$$

we obtain

$$\widehat{\tau}_{N+j} + a(q^j e) + b(q^j e) \le T + \frac{D}{1-q}(1-q^j) + Hq^j < \sigma.$$

This places the first two stages before their remainders are estimated. Then $\widehat{W}_{N+j} \le w(q^j e)$ and $\widehat{C}_{N+j} \le c(q^j e)$ allow the third stage to be placed and the residual to be bounded by $P(q^j e) \le q^{j+1} e$. Induction gives

$$\widehat{\tau}_\infty \le T + \frac{D}{1-q}, \qquad \sum_{i=j}^{\infty}(\widehat{A}_{N+i} + \widehat{C}_{N+i}) \le \frac{Dq^j}{1-q}. \tag{S3.16}$$

This is the uniform estimate, not an extrapolation from a finite sample.

S3.5 Rational states and value of the criterion

The polynomials of the first cycle are evaluated at α_c. All the following comparisons are between exact fractions. We obtain

$$\widehat{V}_0 = 0.292871921615, \quad \widehat{\tau}_1 < 0.198986457 < 1/5, \quad \widehat{\varepsilon}_1 < 0.013187894 < 1/75.$$

We therefore start the upper sequence of the cycles $n = 1, 2$ at $(\overline{\tau}_1, \overline{\varepsilon}_1) = (1/5, 1/75)$. At the end of each cycle, $\mathcal{R}(u) = 10^{-24}\lceil 10^{24}u \rceil$ is applied to the data passed to the next cycle. The exact transmitted states are

$$\begin{aligned} \overline{\tau}_2 &= \frac{23650984109566728133 0063}{10^{24}}, & \overline{\varepsilon}_2 &= \frac{426195191308547945528}{10^{24}}, \\ \overline{\tau}_3 &= \frac{23802613530328471547 8357}{10^{24}}, & \overline{\varepsilon}_3 &= \frac{5818357024425655}{10^{24}}. \end{aligned} \tag{S3.17}$$

The numerators and denominators of all intermediate quantities, not only these four states, are included in `results/criterion_alpha1547.json`.

We take $T = 238026136/10^9$, $e = 6/10^9$, $\sigma = 6/25$, and $q = 1/2$. In this case, $m(\sigma) = 97/361$ and $M(\sigma) = 31250/6859$. The computed fractions satisfy

$$\begin{aligned} P(e)/e &< 6.8 \times 10^{-20} < 1/2, \\ T + 2(a(e) + c(e)) &< 0.238026181 < 0.24, \\ T + a(e) + b(e) &< 0.238026159 < 0.24. \end{aligned} \tag{S3.18}$$

We also verify that

$$(3 - \alpha_c)^2 - 8 = \frac{9573209}{10^8} > 0, \qquad 2(1 - \sigma_{\text{all}})^2 - 1 = \frac{2921}{78125000} > 0.$$

All radii of the cycles $n = 1, 2$ are less than 0.253. Together with the first cycle and (S3.18), this gives $\widehat{\Phi}_{\text{ef}}(\alpha_c) \le \sigma_{\text{all}} < \rho$. Monotonicity extends the result to $0 < \alpha \le \alpha_c$. The hatted version of the semilocal theorem gives, in particular,

$$\|x_* - x_0\| < \frac{0.238026181}{\gamma}, \qquad x_n, y_n, z_n \in B[x_0, 0.24/\gamma] \quad (n \ge 3).$$

For $n = 3 + j$, we also have the bound $\|x_* - x_{3+j}\| \le 2(a(e) + c(e))2^{-j}/\gamma$.

S3.6 Quantitative comparison with Cordero et al. (2015)

We retain the functions of [2]:

$$h_{15}(a) = a/2 + a^2/2 + 5a^3/8, \quad f_{15}(a) = [1 - a(1 + h_{15}(a))]^{-1},$$

$$g_{15}(a) = a/2 + (1 + a)h_{15}(a) + ah_{15}(a)^2/2.$$

The first positive zero of $f_{15}(a)^2 g_{15}(a) - 1$ on the admissible branch satisfies

$$a^*_{15} \in (0.29313992158369, 0.29313992158370).$$

The endpoints are rational; the signs are checked exactly. On the positive branch, h_{15}, f_{15}, g_{15} are increasing, and $f_{15}^2 g_{15}$ is strictly increasing; therefore, the sign change identifies the endpoint of the interval starting from zero.

For $F_\alpha(u, v) = (u + v^2 - \alpha, v + u^2 - \alpha/2)$, with the maximum norm and $x_0 = 0$, the gamma condition uses $\beta = \alpha$, $\gamma = 1$. The smallest constants for the 2015 criterion are $\beta_{15} = 1$, $\eta_{15} = \alpha$, $K = 2$, so $a = 2\alpha$. The admitted interval is $0 < \alpha < a^*_{15}/2 \simeq 0.146569960792$, strictly contained in $0 < \alpha \le 0.1547$. For $\alpha = 3/20$, $a = 3/10$ and $f_{15}(3/10)^2 g_{15}(3/10) > 1.06695 > 1$. Failure of a sufficient condition does not imply divergence of the method.

For the location comparison, we use the unhatted rational controls, started at $(0, \alpha)$, as indicated in the table in this section. Their limit τ_∞ is not identified with $\widehat{\tau}_\infty$. The program `code/compare_cordero2015.py` performs a finite verification for each datum and applies (S3.15) to that sequence. The 2015 bounds for the limit and for all stages are

$$L_{15}(a) = \frac{(1 + h_{15}(a))\alpha}{1 - f_{15}(a)g_{15}(a)}, \qquad R_{15}(a)\alpha = \left(\frac{5a}{2} + \frac{1 + h_{15}(a)}{1 - f_{15}(a)g_{15}(a)}\right)\alpha.$$

The recurrence-based sum for the location is $\sum_{n\ge0}(1 + h_{15}(a_n))d_n$, with $a_0 = a$, $d_0 = \alpha$, $a_{n+1} = a_n f_{15}(a_n)^2 g_{15}(a_n)$, and $d_{n+1} = f_{15}(a_n)g_{15}(a_n)d_n$. Its first two terms sum to

$$Q^{[2]}_{15} = (1 + h_{15}(a))\alpha + (1 + h_{15}(a_1))f_{15}(a)g_{15}(a)\alpha.$$

In all common tabulated cases, fractions are used to check $U_\alpha < Q^{[2]}_{15} < L_{15} < R_{15}\alpha$, where $\tau_\infty \le U_\alpha$. Thus, the improvement does not come from comparing only with a larger geometric majorization.

α	Upper bound for τ_∞	$Q_{15}^{[2]}$	L_{15}
0.01	0.010103200647	0.010310404439	0.010316962789
0.02	0.020427400093	0.021286724646	0.021344357094
0.05	0.053036118831	0.059049322277	0.060285803938
0.1	0.116956870642	0.146445008466	0.166567164179
0.14	0.195079729913	0.262956274107	0.406176058866
0.15	0.230706675248	–	–

Table S3.1: The bound in the first numerical column is rounded upward; the other two columns are approximations. The signs are checked using complete fractions in the program and the JSON file, not these rounded values. The 2015 criterion does not apply at $\alpha = 0.15$.

S3.7 Programs and scope of the verification

`code/verify_gamma_criterion.py` verifies the first cycle, the rational states of the subsequent cycles, and all tail inequalities. `code/compare_cordero2015.py` checks the separation of the criteria and the location bounds. Both verifications use `fractions.Fraction`; they do not decide signs through floating-point arithmetic. Decimals are generated after the comparisons. The justification for all indices is the tail induction, not the execution of a finite number of steps.

S4 Verification of published semilocal analyses

We examine the published results of Cordero et al. [2], Singh et al. [3], and Maroju, Behl, and Motsa [4]. The cited pages are the printed pages of those works. The aim is to clarify which assertions can be used and which corrections their proofs require. No intentions are attributed to the authors, and the falsity of an entire theorem is not inferred from any typographical error.

Four situations are distinguished: an identity or inequality refuted by exact calculation; an insufficient argument; a false conclusion with a counterexample satisfying the assumptions; and a numerical value that does not agree with the published formula. The programs separate rational and symbolic checks from multiprecision reproductions; the latter are not interval-arithmetic results.

S4.1 Equivalence of the method and obstruction to fifth order

In the three articles, the last stage is written as

$$x^+ = z - \tfrac{1}{5}\Gamma(-16F(y) + F(z)), \qquad z = y - 5\Gamma F(y), \qquad \Gamma = F'(x)^{-1}.$$

Substituting z gives $x^+ = y - \Gamma(9F(y) + F(z))/5$, exactly the scheme studied here. The different expressions do not produce different methods. The expansion in Section S1 contains

$$\mathcal{Q}_4(e) = 4\{H_2(e, H_2(e, H_2(e, e))) - H_2(H_2(e, e), H_2(e, e))\}.$$

For $F(u, v) = (u + v^2, v + u^2)$, $H_2(p, q) = (p_2q_2, p_1q_1)$ and $\mathcal{Q}_4(u, v) = 4(v^3 - u^3)(u, -v)$. By (S1.10), $\|T(t, 2t)\|_\infty / \|(t, 2t)\|_\infty^4 \to 7/2$, whereas the quotient with the fifth power diverges. The root is simple and the operator is polynomial. Uniform local fifth order cannot be assigned to the general class. This does not determine a single order for all trajectories or by itself refute semilocal convergence.

S4.2 Cordero et al. (2015): corrections and valid semilocal content

A1. The function h retains the factor of the second stage. On pages 206–207 of [2], the definition is

$$h(a) = a/2 + a^2/2 + 5a^3/8 = \frac{a}{10}\left[4 + (1 + 5a/2)^2\right].$$

This identity is exact. For $F_a(u) = u - 1 - au^2/2$ and $x_0 = 0$, we obtain $y_0 = 1$, $z_0 = 1 + 5a/2$, and $x_1 = 1 + h(a)$. The bound is exact in that example. The missing factor identified in the 2022 work should not be attributed to the 2015 work.

A2. An intermediate inequality in (6) is false. The chain on page 207 includes

$$\|\Gamma_0\| \, \|F(x_0) + (9F(y_0) + F(z_0))/5\| \leq (1 + h(a_0))\|\Gamma_0 F(x_0)\|.$$

Take $F(u, v) = (u, 100v)$, the maximum norm, $x_0 = (0, 1)$, $\Gamma_0 = \mathrm{diag}(1, 1/100)$, $\beta = \eta = 1$, and $K = 1/10$. We have $a_0 = 1/10$ and $y_0 = z_0 = x_1 = 0$. The constant derivative satisfies the Lipschitz bound with this K. The inequality would require $100 \leq 1 + h(1/10) = 1689/1600$, which is false. The necessary bound retains Γ_0 inside the norm and is proved by first performing the Newton cancellation. The example does not refute convergence: the root is reached in the first cycle.

A3. Application of a remainder to a vector. In the proof of (II_1), page 207, a vector appears multiplied on the right by an integral of operators. In Banach spaces, the integral must be written as applied to the vector $-\Gamma_0(9F(y_0) + F(z_0))/5$. The printed identity does not respect the input and output spaces; correcting it preserves the norm bounds.

Reconstruction of the semilocal step. Fix an iteration and set $d = \|\Gamma F(x)\|$, $b \geq \|\Gamma\|$, $\eta \geq d$, and $a = Kb\eta$. If $Q = (9F(y) + F(z))/5$, the remainders from x give

$$F(y) = E_y, \qquad F(z) = -5F(y) + E_z, \qquad Q = (4F(y) + E_z)/5,$$

$$\|E_y\| \leq Kd^2/2, \quad \|E_z\| \leq K\|z - x\|^2/2, \quad \|z - x\| \leq (1 + 5a/2)d.$$

Consequently, $\|\Gamma Q\| \leq h(a)d$ and $\|x^+ - x\| \leq (1 + h(a))d$. The remainder E_+ from y to x^+ gives

$$\Gamma F(x^+) = \Gamma F(y) - \Gamma Q - \Gamma(F'(y) - F'(x))\Gamma Q + \Gamma E_+.$$

Taking norms,

$$\|\Gamma F(x^+)\| \leq [a/2 + (1 + a)h(a) + ah(a)^2/2]d = g(a)d.$$

The variation of the derivative and the Banach lemma give

$$\|\Gamma^+ F'(x)\| \leq f(a), \qquad \|\Gamma^+ F(x^+)\| \leq f(a)g(a)d, \qquad f(a) = [1 - a(1 + h(a))]^{-1}.$$

The published functions are recovered without using a fifth-order assumption. The recurrence $a_{n+1} = a_n f(a_n)^2 g(a_n)$ is decreasing on the interval in Section S3. Formula (8), page 209, retains a geometric tail that tends to zero. The control of z_n retains $(f(a_0)^3 g(a_0)^2)^n$, and the radius includes $5a_0/2$. The dependence between the domain and the inverses is resolved by induction over the stages with these same estimates; the limit belongs to a ball smaller than the one required for the domain.

A4. Uniqueness is indeed proved. Theorem 1 states it, and pages 209–210 use the averaged operator between roots. With the open balls, we obtain

$$K\beta \int_0^1 \|x_* + t(y_* - x_*) - x_0\| \, dt < \frac{K\beta}{2}\left(R\eta + \frac{2}{K\beta} - R\eta\right) = 1.$$

The Banach lemma applies. Uniqueness is not a conclusion missing from that work. The remark on its page 210 discusses the scope of the radii near the threshold.

A5. Order and efficiency indices. Theorem 1 does not prove fifth order, although section 4 uses it to compute indices in finite dimensions. In dimension three, with the costs of that section, generic order four gives $4^{1/35} < 2^{1/17}$; the claimed superiority over Newton for every dimension greater than two is not retained. This does not exclude competitiveness in other problems. Table 2 also computes indices using the total cost, although the definition uses the cost per iteration. For M5, 640 operations per step and four steps produce 2560 operations: the published value 1.000629 corresponds to $5^{1/2560}$, not $5^{1/640}$. The other rows exhibit the same change of convention.

A6. Reproduction of the eight-node example. Gauss–Legendre with 2000 digits gives

$$K \simeq 0.247117984146368, \quad \beta \simeq 1.596943933144693,$$
$$\eta \simeq 0.591064523238323, \quad a_0 \simeq 0.233253900277713.$$

The radii are approximately 1.516495240390509 and 3.551497385109859, in agreement with the published values. Extending the calculation gives ACOC values 4.7225706005, 3.8727238969, 4.0306205065, and 3.9925100905. The residual after four cycles is approximately $1.2044268509 \times 10^{-312}$, less than the published tolerance 10^{-180}; the number of steps is not refuted. The reproduced first component is $1.0122394257\ldots$, compared with $1.01222\ldots$ in their table, a minor discrepancy. The data and the six updates are included in `results/literature_checks/`.

The semilocal core of the 2015 work can be justified with its own functions. The comparison below uses that core, not the incorrect order classification or the efficiency tables.

S4.3 Singh et al. (2016): inequalities, radii, and a local counterexample

B1. Hölder generalization of the auxiliary functions. With $q = 1$, the functions t, r, s on pages 268–269 of [3] coincide with the 2015 functions h, f, g. For $0 < q \le 1$, the estimate of t uses $(1+u)^q \le 1 + u^q$ and does not lose the factor five. The condition $r(c_0)^{1+q} s(c_0)^q < 1$ is consistent with the decrease of c_n.

B2. Incorrect power in the semilocal proof. Relation (V) gives $(1 + t(c_n))d_n$, but (2.17)–(2.20) use $(1 + t(c_n))^q d_n$, which is smaller if $q < 1$. The example

$$F(u) = u - 1 - u^2/2000, \quad x_0 = 0, \quad D = (-1.01, 1.01),$$
$$\beta = \eta = 1, \quad q = 1/2, \quad K = c_0 = 1/700$$

satisfies the semilocal setting. The Hölder condition holds because $(1/1000)^2(2.02) < (1/700)^2$. The auxiliary contraction also holds and the correct radius is less than 1.01; the program checks them with rational bounds for the radicals. The exact stages give $x_1 = 1.000500500625$. If $w = 1/1050$,

$$t(1/700) = w + w^2 + \frac{w(1+5w)}{\sqrt{5250}} < w + w^2 + \frac{w(1+5w)}{72} < \frac{967}{10^6}.$$

It follows that $(1 + t(1/700))^{1/2} < 1.0004835 < 1.000500500625 = x_1$. Inequality (2.17) is false. The radius in Theorem 2.1 uses $1 + t(c_0)$ without the power: removing the incorrect exponents reconstructs the semilocal chain with that radius. The counterexample refutes the printed inequality; it does not prove the semilocal conclusion false.

B3. Local uniqueness: a false conclusion of Theorem 3.1. The necessary condition $K_0 R^q/(1+q) < 1$ is equivalent to $R < ((1+q)/K_0)^{1/q}$. Instead, $R < (1+q)/K_0$ is printed. Consider

$$G(u) = u - 4u^2, \quad x_* = 0, \quad D_0 = (-3/10, 3/10), \quad q = 1/2, \quad K_0 = 9/2, \quad K_1 = 25/4.$$

On the domain, $8|u| \le K_0|u|^{1/2}$ and $8|u-v| \le K_1|u-v|^{1/2}$ hold, since $64(3/10) < (9/2)^2$ and $64(6/10) < (25/4)^2$. Moreover,

$$r_1 = \left(\frac{1+q}{K_1 + (1+q)K_0}\right)^{1/q} = (3/26)^2 < 7/25.$$

The local radius ρ used here is less than r_1. The choice $R = 7/25$ satisfies $\rho < R < (1+q)/K_0 = 1/3$ and $B[0,R] \subset D_0$, but contains the roots 0 and $1/4$. Uniqueness in that ball is false. The corrected bound is $1/9$ and excludes the second root. The refutation specifically concerns local uniqueness with the printed radius.

B4. Radii of the semilocal examples. The expressions for η have typographical errors: in Example 2.1, 16 appears where 24 is required in the denominator; in Example 2.2, a factor $1/4$ is missing from the numerator. Recalculating from the operators and using the formula in Theorem 2.1 gives: For $q < 1$, the published existence radii are reproduced with the incorrect

Case	Published existence	Recomputed	Published uniqueness	Recomputed
2.1, $q = .85$	.04630	.04632668	38.25348	38.11497803
2.1, $q = 1$	.04620	.04620900	21.99790	21.95379100
2.2, $q = .7$	.47873	.49097189	6.85383	5.89723102
2.2, $q = 1$	.44916	.44915651	3.65166	3.32162365

Table S4.1: Multiprecision reproduction of the 2016 radii; these are not rigorous intervals.

power $(1 + t(c_0))^q$. The uniqueness radii are reproduced using

$$\eta\left(\frac{1+q}{c_0} - (R\eta)^q\right)^{1/q} \quad \text{instead of} \quad \eta\left(\frac{1+q}{c_0} - R^q\right)^{1/q}.$$

The scale η is applied twice. This does not exclude justifying some of the radii by another argument; it indicates that they do not agree with the theorem's formula. For Example 2.3, the initial vector yields $\beta \simeq 26.5887634801$ and $\eta \simeq 0.000375702982175$.

B5. Local examples and table. In Example 3.1, zero is also a root by the definition of G; $x_* = 1$ is not the only root in the entire stated interval. This does not exclude local uniqueness near 1. In Example 3.2, $G(u) = 2u^{3/2}/3 - u$, $x_* = 9/4$, and $G'(x_*) = 1/2$. The constant $K_1 = 1$, with $q = 1/2$, does not control all of $[1,3]$, since $2(\sqrt{3}-1) > \sqrt{2}$; the inequality persists at interior points sufficiently close to the endpoints. The formulas give $r_1 = .36$, not .361. For $q = 1/2$, the local radii of Examples 3.1–3.4 are reproduced as $1.291615219 \times 10^{-6}$, 0.01206846112, 0.00194510607, and $6.843277886 \times 10^{-5}$. Table 2 assigns the last three values to other rows. Fractional powers require an explicit real domain; varying q in x^{1+q} also changes the equation, not only the theory applied to a fixed problem.

B6. Recoverable content. With $H = 1 + t(c_0)$ and $C = r(c_0)s(c_0) < 1$, the corrected recurrence gives

$$\|x_{n+m} - x_n\| \le H\eta C^n \frac{1 - C^m}{1 - C}.$$

The stages are controlled by $H\eta/(1-C)$ and $5c_0C^n\eta/(1+q)$, which explains the radius in the statement without the incorrect power. An induction over the stages avoids circularity between the domain and the inverses. In semilocal uniqueness, the open balls and $q > 0$ allow the strict inequality to be recovered. The objection concerning merely nondecreasing continuity in the 2022 work does not automatically transfer to this case.

S4.4 Maroju, Behl, and Motsa (2022): semilocal and local flaws

C1. Incompatibility of C6 and the recurrence. Page 23 of [4] requires $\varphi \in C[0,1]$, $\varphi \geq 1$, and defines $a_{n+1} = a_n f(a_n)\varphi(c_n)$, $c_n = f(a_n)g(a_n)$. When $a_n > 0$, $f(a_n) > 1$, and $c_n \in [0,1]$, we have $a_{n+1} > a_n$, contrary to Lemma 2.2. If c_n lies outside that interval, the assumption does not define $\varphi(c_n)$. The auxiliary functions also use $\varphi(1+h(a))$ and $\varphi(1+a_0M)$ outside the domain. The condition $f(a_0)\varphi(a_0) < 1$ is also incompatible in the stated regime. Changing a sign without reconstructing the domain and scaling does not resolve these dependencies.

C2. Missing factor five and false bound (2.6). The estimates on page 24 give $\|z_0 - x_0\| \leq (1+5a_0M)\|y_0 - x_0\|$, but (2.6) uses $1 + a_0M$. For $F(u) = u - 1 - u^2/20$, $x_0 = 0$, $\beta = \eta = 1$, and $\omega(t) = t/10$, extend $\varphi(s) = \max\{1, s\}$ to $s \geq 0$ to isolate that inequality. C6 is retained on $[0,1]$, and $M = \int_0^1 \varphi(t)dt = 1$. The stages are $y_0 = 1$, $z_0 = 5/4$, and $a_0 = 1/10$, but

$$\omega(|z_0 - x_0|)|z_0 - x_0| = \frac{5}{32} > \frac{121}{1000} = \omega(\eta)\varphi(1+a_0M)(1+a_0M)\eta.$$

The bound is refuted even when only the domain of φ is repaired. No claim is made that the example satisfies the incompatible contraction conditions in C1.

C3. The function h underestimates the displacement. If C6 is repaired as in the Lipschitz case, with $\varphi(s) = s$ and $M = 1/2$, then for $F_a(u) = u - 1 - au^2/2$ we obtain

$$x_1 = 1 + a/2 + a^2/2 + 5a^3/8, \qquad h_{\text{imp}}(a) = a/2 + a^2/10 + a^3/40.$$

Hence $x_1 - (1 + h_{\text{imp}}(a)) = a^2(3a+2)/5 > 0$. At $a = 1/10$, (2.7) would require $1.055625 \leq 1.051025$. The choice $\varphi(s) = s$ does not literally satisfy C6: it shows that this isolated repair does not save the bound.

C4. Residual propagation. The definition of h enters (2.11)–(2.12), which are intended to control $\|\Gamma_1 F(x_1)\|$. An estimate compatible with the second stage would need to retain at least

$$h_{\text{corr}}(a) = \tfrac{4}{5}aM + \tfrac{1}{5}aM\varphi(1+5aM)(1+5aM).$$

The form $g(a) = aM + (1+a)h(a) + ah(a)M\varphi(h(a))$ may be a reasonable bound, but it requires a valid h and appropriate scaling conditions. The missing factor affects the chain that must prove the decay of the normalized residual.

C5. Mixing recurrences and the Cauchy tail. Relation (II) gives factors $f(a_j)g(a_j)$ for $\|\Gamma_n F(x_n)\|$, not $f(a_0)\varphi(c_0)$, which belongs to another recurrence. The substitution in (2.15)–(2.17) mixes these quantities. The sum in (2.17) starts at zero and does not retain the tail that must tend to zero with its initial index. An appropriate bound would be of the form $C\eta c_0^m(1 - c_0^n)/(1 - c_0)$, after proving the contraction. The published bound does not by itself prove the Cauchy property.

C6. Containment and domain. In (2.19), $1 + h(a_0)$ is replaced by $\varphi(1 + h(a_0))$ without an identity that permits it. The last inequality in (2.21) requires $5Ma_0 + R(1 - c_0^{n+1}) < R$. If $0 < c_0 < 1$ and $Ma_0 > 0$, the left-hand side tends to $5Ma_0 + R > R$: the estimate does not hold for every index. The statement does not clearly include containment of the full ball in the domain, and derivatives are used before containment is established. The induction needs to be reconstructed, not just a constant changed.

C7. Mixed regularity assumptions. Theorem 2.3 introduces $a_0 = K\beta\eta^p$ and uses a Hölder bound for $F'(x_n)$, although the preceding framework uses $a_0 = \beta\omega(\eta)$ and general ω-continuity. The consistent bound would be $\|F'(x_n)\| \le \|F'(x_0)\| + \omega(R\eta)$. The identity $F(x_n) = F'(x_n)\Gamma_n F(x_n)$ is correct; the decay of the second factor remains to be justified with the printed recurrences.

C8. Semilocal uniqueness. Formula (2.22) reaches a bound ≤ 1 and invokes the Banach lemma, which requires a strict inequality. Nondecreasing continuity of ω does not guarantee this merely by using open balls. Also, $R + r$ appears where the distances are bounded by $R\eta + r$. A consistent sufficient condition would be

$$\beta \int_0^1 \omega((1-t)R\eta + tr)\, dt < 1,$$

with the domain inclusions. It is not claimed that uniqueness is false in every example, but that the printed argument does not prove it.

C9. Identities and coefficients in the local part. Formula (3.14) represents $F(y_0)$ by the segment ending at x_0, whose integral is $F(x_0) - F(x_*)$, not $F(y_0)$; y_0 must be used. There is a similar difficulty in (3.18). In (3.19), the coefficient 16 disappears and expressions involving $F'(x_*)F(y_0)$ appear where the inverse is required to make the spaces compatible.

This does not imply that the printed function g_3 cannot be a bound. Writing $D = 1 - \omega_0(t)$ and $A_i = \int_0^1 [1 + \omega_0(\theta g_i(t)t)]d\theta$, we have $g_2 = g_1 + 5A_1g_1/D$. The combined stage allows $g_3^* = g_1 + (9A_1g_1 + A_2g_2)/(5D)$, whereas the printed formula, interpreted with the missing parenthesis, is $g_3 = g_2 + (A_1g_1 + A_2g_2)/(5D)$. Hence $g_3 - g_3^* = 17A_1g_1/(5D) \ge 0$; a bound can be recovered by another argument without validating the published identity. The uniqueness operator on page 31 integrates between x_* and $2x_* - y_*$, not between the two roots. One must use $\int_0^1 F'(y_* + \theta(x_* - y_*))d\theta$. The construction of radii does not cover $\omega_0 = \omega = 0$, when $g_1 = 0$ and $g_1 - 1$ has no positive zero. An infinite-radius convention or a region where $g_i < 1$ is needed.

C10. Local example 3.2 does not satisfy the equation. Its operator satisfies $F(0) = -1$, although $x_* = 0$ is identified as a root. The Green kernel vanishes at the endpoints, so every solution satisfies $x(0) = x(1) = 1$ and does not belong to a ball centered at zero with radius 0.224209. The real power $x^{3/2}$ is also not defined on any full open ball around the zero function. In $F'(x)u$, the factor $u(t)$ is missing under the integral. The printed functions give

$$r_0 = \left(\frac{\sqrt{137} - 3}{4}\right)^2 \simeq 4.73573753348,$$

not 3.2000. The values $r_1 \simeq 2.63030003815$ and $r_2 \simeq 0.448599953403$ are reproduced. For the printed g_3, we obtain $r_3 \simeq 0.336536461475$; 0.224209 corresponds to restoring the coefficient 16. No adjustment of these radii resolves $F(0) \neq 0$.

C11. Domain of semilocal example 2.5. The constants $\beta = 40/31$, $\eta = 11/62$, and $a_0 \simeq 0.137224097318$ are consistent with the initial function 1, but no φ completing the assumptions is specified. If $u^{8/5} = |u|^{8/5}$ is interpreted on the whole real space,

$$\|F'(x) - F'(y)\| \le (\|x - y\|^{3/5} + \|x - y\|)/5$$

fails for the constant functions $x = .01$, $y = -.01$: the two sides are approximately 0.0257382937792 and 0.0231270499958. Restricting to the positive cone avoids that example, but requires an open domain and its verification. On a set of diameter D,

$\omega(t) = (t^{3/5} + t)/5 \le (1 + D^{2/5})t^{3/5}/5$ for $0 \le t \le D$. One cannot claim without qualification that every Hölder condition fails; on a ball separated from zero, the derivative can be Lipschitz.

The particular conclusions admit another check. For $Tx = 1 + \mathcal{K}(|x|^{8/5} + x^2/10)$, $\|\mathcal{K}\| = 1/8$, on $B[1, 0.238277]$, the approximate bounds are $\|Tx - 1\| \le 0.19512846754$ and Lipschitz constant $0.25832039011 < 1$. On $B[1, 2.54114]$, one constant is $0.51561500993 < 1$. `code/verify_maroju_bounds.py` verifies with fractions $\|Tx - 1\| < 0.196 < 0.238277$ and constants less than 0.259 and 0.516. The fractional powers are enclosed by comparing integer powers. This justifies those particular conclusions by a fixed-point argument, not the general proof or the containment of the method's stages.

C12. Order under weak regularity. The local bounds g_i are $O(t)$ in the Lipschitz case and give, through that route, a quadratic bound, not fifth order. Under Hölder conditions, they give an exponent $1+q$. Even in dimension one, $F(u) = u+u|u|$ has a Lipschitz derivative and a simple root, but from $t > 0$ the scheme produces $T(t) = 32t^4/5 + O(t^5)$. Its lack of higher regularity matters. Regularity, local order of a smooth scheme, and order of a semilocal majorization are different assertions.

S4.5 Assessment and relation to the quantitative comparison

The defects in the 2015 work are corrected while retaining its functions and semilocal mechanism, including uniqueness. In the 2016 work, the incorrect power invalidates printed steps of the proof, but the semilocal radius in the statement admits a reconstruction; the counterexample in B3 refutes local uniqueness with the published radius. In the 2022 work, the incompatibilities of the assumptions and the chain of bounds require a more substantial reconstruction. In none of these cases does the designation "fifth order" replace an independent proof for operators in Banach spaces.

The comparison in Section S3 uses the corrected semilocal criterion of the 2015 work, not a weakened version. It distinguishes the radius for all stages, the location of the limit, and the recurrence-based sum. The advantage is proved on a common quadratic family and is not inferred from errors found in other works. The analysis is limited to the results examined here.

S5 Applications: derivatives, histories, and reproducibility

The five problems use the parameters specified below and are formulated in $C[0, 1]$ with the uniform norm. The gamma conditions are checked for the continuous operators. The following histories come from solving their discretizations with $N = 30$, five cycles, and 1000 digits. For Chandrasekhar and Bratu, $N = 20, 40, 60$ are also included with the same precision and number of cycles. The eleven calculations and profiles are reproduced through `code/reproduce_experiments.py`. These multiprecision floating-point results are not identified with the rational checks in Section S3.

S5.1 Continuous verification and parameters of the five problems

In Problems 1–4 we take $\Omega = C[0, 1]$; in the fifth, a positive domain is specified. By the Lipschitz form of the gamma difference condition established in Section S3, in addition to $\|G(x_0)\| \le \beta$ and $B[x_0, R_\gamma] \subset \Omega$, it suffices to verify

$$\|G''(x_0)\| \le 2\gamma, \qquad \|G''(x) - G''(z)\| \le L\|x - z\|_\infty, \qquad L \le 6\gamma^2$$

on $B[x_0, R_\gamma]$, and check $\alpha = \beta\gamma \le \alpha_c = 1547/10000$. Products, quotients, and exponentials of functions are understood pointwise. For Problems 1 and 5 we define

$$(\mathcal{K}_0 x)(s) = \int_0^1 K_0(s, t)x(t)\, dt, \qquad K_0(s, t) = \begin{cases} s/(s + t), & s > 0, \\ 0, & s = 0. \end{cases}$$

Although K_0 is not continuous at $(0,0)$, $\mathcal{K}_0$ maps $C[0,1]$ into itself: continuity for $s > 0$ is immediate, and

$$|(\mathcal{K}_0 x)(s)| \le \|x\|_\infty s \log \frac{1+s}{s} \longrightarrow 0 \quad (s \to 0^+).$$

Moreover, $\|\mathcal{K}_0\| = \log 2$.

Problem 1. Chandrasekhar's integral equation. For the family in [5], we take $\omega_0 = 1/2$, $x_0 = 1$, and

$$\begin{aligned} F(x) &= x - 1 - \tfrac{\omega_0}{2} x\mathcal{K}_0 x, \\ F'(x)h &= h - \tfrac{\omega_0}{2}(h\mathcal{K}_0 x + x\mathcal{K}_0 h), \\ F''(x)[h,k] &= -\tfrac{\omega_0}{2}(h\mathcal{K}_0 k + k\mathcal{K}_0 h). \end{aligned}$$

With $q = \omega_0 \log 2 < 1$, we have $\|I - F'(1)\| \le q$, $\|F'(1)^{-1}\| \le (1-q)^{-1}$, $\|F(1)\|_\infty \le q/2$, and $\|F''(1)\| \le q$. We choose

$$\beta = \gamma = \frac{q}{2(1-q)} \simeq 0.265197109517, \qquad \alpha \simeq 0.0703295068963 < \alpha_c.$$

Thus, $\|G(1)\| \le \beta$ and $\|G''(1)\| \le 2\gamma$. Since G'' is constant, $L = 0$.

Problem 2. Polynomial Hammerstein equation. We use the equation in [6, Example 2, equation (3.3), p. 4617]:

$$\begin{aligned} [F(x)](s) &= x(s) - \sin(s/3) - s\int_0^1 t^4 (x(t)^3 + x(t)^2/2)\, dt, \\ [F'(x)h](s) &= h(s) - s\int_0^1 t^4 (3x(t)^2 + x(t))h(t)\, dt, \\ [F''(x)[h,k]](s) &= -s\int_0^1 t^4 (6x(t)+1)h(t)k(t)\, dt. \end{aligned}$$

For $x_0 = 0$, $F'(0) = I$, $\|G(0)\|_\infty = \sin(1/3)$, and $\|G''(0)\| \le 1/5$. Moreover, $\|G''(x) - G''(z)\| \le (6/5)\|x - z\|_\infty$. Therefore, we take

$$\beta = \sin(1/3), \qquad \gamma = 1/\sqrt{5}, \qquad \alpha = \frac{\sin(1/3)}{\sqrt{5}} \simeq 0.146325916783 < \alpha_c,$$

with $L = 6/5 = 6\gamma^2$ and $\|G''(0)\| < 2\gamma$.

The rank-one structure allows a scalar reduction. Let $\phi(s) = \sin(s/3)$, $\psi(s) = s$, and $\mathcal{L} = \{x_a = \phi + a\psi : a \in \mathbb{R}\}$. Define

$$h(a) = a - \int_0^1 t^4 \left[(\phi(t) + a\psi(t))^3 + \tfrac{1}{2}(\phi(t) + a\psi(t))^2\right] dt.$$

Then $F(x_a) = h(a)\psi$ and $F'(x_a)\psi = h'(a)\psi$. If $F'(x_a)$ is invertible, $h'(a) \neq 0$ and $F'(x_a)^{-1}F(x_b) = h(b)\psi/h'(a)$ for $x_b \in \mathcal{L}$. Since $F'(0) = I$, the first step gives $y_0 = \phi$, and also $z_0, x_1 \in \mathcal{L}$. From $n = 1$ onward, the three stages coincide with the scalar method for $h(a) = 0$. Near a simple root, the order is at least five, and exactly five when the leading coefficient does not vanish. Since F'' takes values in $\operatorname{span}\{\psi\}$ and $F'(x_*)^{-1}$ preserves that direction, the errors on the invariant affine line also satisfy $e \cdot e \in \operatorname{span}\{e\}$ for $u \cdot v = \frac{1}{2}F'(x_*)^{-1}F''(x_*)[u,v]$. Thus the fourth-degree term vanishes along this sequence, as in Proposition 2.4 of the article. Quadrature preserves rank one and this reduction.

Problem 3. Bratu through Green's function. For $x'' + \lambda e^x = 0$, $x(0) = x(1) = 0$, we take $\lambda = 2$. Its integral formulation uses

$$(\mathcal{K}x)(s) = \int_0^1 K(s,t)x(t)\, dt, \qquad K(s,t) = \begin{cases} t(1-s), & t \le s, \\ s(1-t), & s \le t, \end{cases}$$

with $\|\mathcal{K}\| = 1/8$; see [7]. The expressions for F and its derivatives are

$$F(x) = x - \lambda\mathcal{K}(e^x), \quad F'(x)h = h - \lambda\mathcal{K}(e^x h), \quad F''(x)[h,k] = -\lambda\mathcal{K}(e^x hk).$$

For $x_0 = 0$, $\|F'(0)^{-1}\| \le (1-\lambda/8)^{-1}$ and $\|G(0)\|, \|G''(0)\| \le \lambda/(8-\lambda) = 1/3$. We choose $\beta = 1/3$ and $\gamma = 0.4$, so that $R_\gamma \simeq 0.732233047034$ and

$$L = \frac{\lambda e^{R_\gamma}}{8-\lambda} \simeq 0.693239845939 < 6\gamma^2 = 0.96, \qquad \alpha = \beta\gamma = \frac{2}{15} < \alpha_{\mathrm{c}}.$$

The Lipschitz bound follows from $|e^{x(t)} - e^{z(t)}| \le e^{R_\gamma}|x(t) - z(t)|$ in the ball.

Problem 4. Hammerstein with a nonseparable kernel. As a specialization of the classes in [8], let $(\mathcal{K}_{\mathrm{H}}x)(s) = \int_0^1 e^{st}x(t)\,dt$, $\lambda = 10^{-2}$, and

$$[F(x)](s) = x(s) - \sin(s/3) - \lambda[\mathcal{K}_{\mathrm{H}}(e^x - 1 - x)](s).$$

We have $\|\mathcal{K}_{\mathrm{H}}\| = e - 1$, $F'(x)h = h - \lambda\mathcal{K}_{\mathrm{H}}((e^x - 1)h)$, and $F''(x)[h,k] = -\lambda\mathcal{K}_{\mathrm{H}}(e^x hk)$. For $x_0 = 0$, $F'(0) = I$. We take $\beta = \sin(1/3)$ and $\gamma = 0.2$, with $R_\gamma \simeq 1.46446609407$. Then

$$\|G''(0)\| \le \lambda(e-1) < 0.4 = 2\gamma, \qquad L = \lambda(e-1)e^{R_\gamma} \simeq 0.0743196987942 < 0.24 = 6\gamma^2,$$

and $\alpha = \beta\gamma \simeq 0.0654389393592 < \alpha_{\mathrm{c}}$.

Problem 5. Reciprocal Hammerstein equation. The reciprocal equation in [9, equation (6), p. 2941] has the same kernel and nonlinearity with coefficient $1/4$; here we use $\mu = 0.1$ and $x_0 = 1$. The numerical protocol is the one specified in this section, not a reproduction of the cited experiment. On the open convex set $\Omega = \{x \in C[0,1] : \min_{t\in[0,1]} x(t) > 0\}$, we define

$$F(x) = x - 1 + \mu\mathcal{K}_0(1/x), \qquad \mu = 0.1,$$
$$F'(x)h = h - \mu\mathcal{K}_0(h/x^2), \qquad F''(x)[h,k] = 2\mu\mathcal{K}_0(hk/x^3).$$

With $x_0 = 1$, we choose $\beta = \mu\log 2/(1 - \mu\log 2) \simeq 0.0744770755493$ and $\gamma = 0.8$. The Banach lemma gives $\|G(1)\| \le \beta$ and $\|G''(1)\| \le 2\beta < 2\gamma$. Since $R_\gamma \simeq 0.366116523517 < 1$, the gamma ball is contained in Ω. With $\delta = 1 - R_\gamma$, the inequality $|x(t)^{-3} - z(t)^{-3}| \le 3\delta^{-4}|x(t) - z(t)|$ gives

$$L = \frac{6\mu\log 2}{\delta^4(1 - \mu\log 2)} \simeq 2.76780915206 < 3.84 = 6\gamma^2.$$

Therefore, $\alpha = \beta\gamma \simeq 0.0595816604395 < \alpha_{\mathrm{c}}$.

S5.2 Discretization, linear solve, and stopping criterion

Let $(t_j, w_j)_{j=1}^N$ be the Gauss–Legendre nodes and weights on $[0,1]$, computed with the same precision as the iterations. The Nyström matrix is defined by $K_{ij} = w_j K(t_i, t_j)$, with the kernel corresponding to each problem. For the kernel K_0, the point $(0,0)$ is not evaluated, since the Gauss–Legendre nodes are interior. For the polynomial problem, the coefficients $w_j t_j^4$ are used and the rank-one structure of the discrete system is preserved exactly. The Jacobians are evaluated using the preceding analytical formulas, not finite differences.

In each cycle, the Jacobian $A = F_N'(x_n)$ is factored only once: $PA = LU$. The three right-hand sides are $F_N(x_n)$, $F_N(y_n)$, and $9F_N(y_n) + F_N(z_n)$. For each one, $Lv = Pb$ and $Uu = v$ are solved by reusing P, L, U. The code also records the numerical residual of these linear equations. This check differs from a proof of backward stability.

We use multiprecision floating-point arithmetic from `mpmath`, with 1000 decimal working digits; no double-precision binary parameters or nodes from a lower-precision rule are introduced. Exactly five cycles are executed, regardless of the residual. If $d_n = \|x_n - x_{n-1}\|_\infty$, we compute

$$\widehat{p}_n = \frac{\log(d_n/d_{n-1})}{\log(d_{n-1}/d_{n-2})}, \qquad n \ge 3.$$

The quotient is omitted if any of the three increments is zero, if the denominator is zero, or if an increment does not exceed the precautionary level $10^{-D+30}\max\{1, \|x_n\|_\infty\}$, where D is the decimal precision. The same comparison marks residuals near the precision limit; it does not turn them into a rigorous bound of 10^{-D}.

S5.3 Profile reconstruction and Bratu reference

To compare different dimensions, nodal vectors are not subtracted directly. In Chandrasekhar, we reconstruct

$$u_N(s) = \left(1 - \frac{1}{4}\sum_{j=1}^{N} w_j \frac{s}{s+t_j} x_{N,j}\right)^{-1} \qquad (s>0), \qquad u_N(0) = 1, \tag{S5.1}$$

and in Bratu,

$$u_N(s) = 2\sum_{j=1}^{N} w_j K(s,t_j) e^{x_{N,j}}. \tag{S5.2}$$

These expressions define the Nyström discretizations used in the computations. The profiles are evaluated with 100 digits on $\mathcal{G} = \{i/400 : 0 \le i \le 400\}$.

The Bratu reference is

$$u_*(s) = 2\log\frac{\cosh(a/2)}{\cosh(a(s-1/2))}, \qquad \frac{2a^2}{\cosh(a/2)^2} = 2, \quad 0<a<2. \tag{S5.3}$$

Indeed, $u_*(0) = u_*(1) = 0$, $u_*''(s) = -2a^2\operatorname{sech}^2(a(s-1/2))$, and $e^{u_*(s)} = \cosh^2(a/2)\operatorname{sech}^2(a(s-1/2))$; the relation for a gives $u_*'' + 2e^{u_*} = 0$. The function $2a^2/\cosh^2(a/2)$ is strictly increasing on $(0,2)$, since its logarithmic derivative is $2/a - \tanh(a/2) > 0$. The chosen root corresponds to the small branch and is computed with 100 digits.

For Chandrasekhar, we report $\max_{s\in\mathcal{G}} |u_N(s) - u_{60}(s)|$; for Bratu, $\max_{s\in\mathcal{G}} |u_N(s) - u_*(s)|$. These are maxima on a finite grid. The first uses a numerical reference, and neither constitutes a bound for the continuous supremum. The change of slope of the Green kernel along $s = t$ is retained in the global quadrature used; spectral convergence of that rule is not assumed.

S5.4 Complete histories for $N = 30$

The tables show the six states $x_0, \dots, x_5$. The sign † indicates that the residual reached the precautionary precision level and is not interpreted as an exact bound. The ACOC of the fifth update uses the preceding increments, which do remain above that level. The vectors of all stages and the values without presentation rounding are included in the JSON and CSV files.

n	$\|F_N(x_n)\|_\infty$	d_n	$\widehat{p}_n$
0	1.732117×10^{-1}	–	–
1	1.583097×10^{-6}	2.51117×10^{-1}	–
2	7.980421×10^{-27}	2.358968×10^{-6}	–
3	$1.832861 \times 10^{-111}$	9.793127×10^{-27}	4.054341706
4	$1.486176 \times 10^{-450}$	$1.991172 \times 10^{-111}$	4.15526647
5	†	1.80824×10^{-450}	4.003242371

Table S5.1: Chandrasekhar: $N = 30$, five cycles, and 1000 digits.

n	$\|F_N(x_n)\|_\infty$	d_n	$\widehat{p}_n$
0	3.267054×10^{-1}	–	–
1	8.26069×10^{-6}	3.400329×10^{-1}	–
2	3.491167×10^{-28}	9.104126×10^{-6}	–
3	$4.706249 \times 10^{-140}$	3.847615×10^{-28}	4.893408432
4	$2.095062 \times 10^{-699}$	$5.186758 \times 10^{-140}$	5.000002792
5	†	$2.308968 \times 10^{-699}$	5.0

Table S5.2: Polynomial Hammerstein: $N = 30$, five cycles, and 1000 digits.

n	$\|F_N(x_n)\|_\infty$	d_n	$\widehat{p}_n$
0	2.497788×10^{-1}	–	–
1	2.523014×10^{-6}	3.288667×10^{-1}	–
2	2.896231×10^{-26}	3.553876×10^{-6}	–
3	$1.165631 \times 10^{-105}$	3.38339×10^{-26}	4.031427609
4	$1.329004 \times 10^{-423}$	$1.620031 \times 10^{-105}$	3.961762207
5	†	$1.549674 \times 10^{-423}$	4.009328914

Table S5.3: Bratu–Green: $N = 30$, five cycles, and 1000 digits.

n	$\|F_N(x_n)\|_\infty$	d_n	$\widehat{p}_n$
0	3.267054×10^{-1}	–	–
1	4.771053×10^{-10}	3.271316×10^{-1}	–
2	1.276932×10^{-45}	4.787416×10^{-10}	–
3	$1.467877 \times 10^{-187}$	1.279543×10^{-45}	4.026550214
4	$1.155496 \times 10^{-755}$	$1.472923 \times 10^{-187}$	3.990068027
5	†	$1.157858 \times 10^{-755}$	4.002458988

Table S5.4: Nonseparable Hammerstein: $N = 30$, five cycles, and 1000 digits.

n	$\|F_N(x_n)\|_\infty$	d_n	$\widehat{p}_n$
0	6.928469×10^{-2}	–	–
1	4.112001×10^{-10}	7.287908×10^{-2}	–
2	2.875395×10^{-44}	4.298189×10^{-10}	–
3	$9.334016 \times 10^{-179}$	2.963562×10^{-44}	4.151191393
4	$7.602065 \times 10^{-719}$	9.75626×10^{-179}	3.936672652
5	†	$7.835002 \times 10^{-719}$	4.016099651

Table S5.5: Reciprocal Hammerstein: $N = 30$, five cycles, and 1000 digits.

Problem	N	$\widehat{p}_5$	Difference on $\mathcal{G}$
Chandrasekhar	20	4.003222946	$5.697734303 \times 10^{-5}$
Chandrasekhar	30	4.0032423709	$8.08036804 \times 10^{-6}$
Chandrasekhar	40	4.0029442845	$1.100911601 \times 10^{-6}$
Chandrasekhar	60	4.0031870353	reference
Bratu–Green	20	4.0098824137	$1.770558072 \times 10^{-3}$
Bratu–Green	30	4.0093289144	$7.862010152 \times 10^{-4}$
Bratu–Green	40	4.00912517	$4.431748795 \times 10^{-4}$
Bratu–Green	60	4.008982179	$1.990539703 \times 10^{-4}$

Table S5.6: Maxima on the grid of 401 points. The reference is $N = 60$ for Chandrasekhar and the explicit small branch for Bratu.

S5.5 Refinement of Chandrasekhar and Bratu

The following table is computed using the reconstructions (S5.1)–(S5.2). All iterations use 1000 digits; only the evaluation of the profiles and the Bratu reference uses 100 digits. The rounded values in this table agree with the stored histories and profiles. The observed stability of the ACOC and reduction of the profile differences do not replace a rigorous analysis of the Nyström error.

S5.6 Files, environment, and scope

The directory `results/experiments/` contains eleven subdirectories, one per problem, number of nodes, and precision. Each includes `result.json` (parameters, environment, and summary), `history.csv`, `nodes_weights_solution.csv`, and `complete_iterates.json`. The latter also contains the intermediate stages of the five cycles. The aggregate files are `five_problems_N30.csv`, `refinement.csv`, `profiles_grid401.csv`, and `Bratu_reference.json`.

The included executions use Python 3.13.5, `mpmath` 1.3.0 and its Python backend, on Linux x86_64. Each record stores the platform, the full version, and the SHA-256 hash of the program. No performance conclusions are drawn from the times, since the hardware conditions and system load were not controlled.

The programs allow the observations for the discrete system to be reproduced, but do not certify rounding errors, quadrature errors, or errors in the continuous supremum. Such validation would require additional bounds or interval arithmetic. The conclusions for the exact iteration in $C[0,1]$ follow from the verified assumptions and the semilocal criterion.

S6 Methodological distinction from previous perturbed majorizations

We document the comparison with Argyros [10] and Ferreira–Svaiter [11]. The formulas from those sources are distinguished from the algebraic correspondences we establish for the scheme studied here. The comparison is limited to the identified constructions and results; the letters used to describe each previous work belong to that context and are not automatically identified with our controls.

S6.1 The framework of Argyros includes exact corrections

On pages 120 and 132 of [10], a Newton stage and a Newton-like stage, respectively, are followed by a correction. Calling this correction d_n, the structure is

$$y_n = x_n - A(x_n)^{-1}F(x_n), \qquad x_{n+1} = y_n - d_n.$$

The author includes $A(x_n) = F'(x_n)$ and exact choices of d_n. It is not appropriate to describe that entire framework as an analysis of external computational errors. Section II, page 121, incorporates an additive term α_n into the scalar remainder:

$$t_{n+1} = s_n + I_n, \qquad h_{n+1} = \frac{L}{2}(t_{n+1} - s_n)^2 + \alpha_n, \qquad s_{n+1} = t_{n+1} + \frac{h_{n+1}}{1 - Lt_{n+1}}.$$

Accumulated bounds for $\sum \alpha_i$ and $\sum I_i$ are required. In the application in its Section VI, page 136, it provides

$$\alpha_n = \frac{N}{6}(s_n - t_n)^3 + M(s_n - t_n)(t_{n+1} - s_n),$$

with derivative control constants. This is an explicit precedent for additive terms constructed with scalar increments for an exact correction. Its Remark 1(a), page 123, also uses radial functions instead of a fixed Lipschitz constant.

Correspondence with our scheme. This observation is a deduction from the comparison, not an assertion in [10]. For our method, let

$$Q_n = \frac{9F(y_n) + F(z_n)}{5}, \qquad \Gamma_n = F'(x_n)^{-1}, \qquad d_n = \Gamma_n Q_n.$$

Then $x_{n+1} = y_n - d_n$ formally has the preceding structure. This does not verify the assumptions on majorants, sums, and domains of Argyros's theorem, nor does it automatically locate the additional point z_n.

The relation also appears in the remainder. Set $p_n = y_n - x_n$, $q_n = z_n - y_n$, $r_n = x_{n+1} - y_n$, $D_n = F'(y_n) - F'(x_n)$,

$$\mathcal{B}_n(d) = F(y_n + d) - F(y_n) - F'(y_n)d, \qquad w_n = r_n - q_n/5.$$

The term in condition (4) of [10], evaluated at (x_n, y_n, x_{n+1}), contains

$$F(y_n) - F(x_n) - F'(x_n)(y_n - x_n) + F'(y_n)r_n = F(y_n) + F'(y_n)r_n,$$

because the Newton step cancels $F(x_n) + F'(x_n)p_n$. For the scheme studied, the residual identity gives

$$F(y_n) + F'(y_n)r_n = D_n w_n - \tfrac{1}{5}\mathcal{B}_n(q_n).$$

Its estimate can use the cancellations in (S3.4). The distinction is not the prior absence of an additive remainder, but its specific control together with the location of the three stages and the comparison with the fixed scalar trajectory.

S6.2 The Ferreira–Svaiter slack is not the realized residual

In sections 4 and 5 of [11], a step s from x with tolerance $\theta \geq 0$ satisfies

$$\|F'(x_0)^{-1}(F(x) + F'(x)s)\| \leq \theta \|F'(x_0)^{-1}F(x)\|.$$

The analysis introduces

$$K(t, \epsilon) = \{x : \|x - x_0\| \leq t, \quad \|F'(x_0)^{-1}F(x)\| \leq f(t) + \epsilon\}$$

and, in formula (4.3) of that source,

$$t^+ = t - (1 + \theta)\frac{f(t) + \epsilon}{f'(t)}, \qquad \epsilon^+ = \epsilon + 2\theta(f(t) + \epsilon). \tag{S6.1}$$

The relation $N_\theta(K(t, \epsilon)) \subset K(t^+, \epsilon^+)$ is the invariance mechanism of its Proposition 4.5. It is not merely a similarity in notation with $g(t_n) + \widehat{\Delta}_n$: there is a coupled construction of a radius and a residual slack.

The variable ϵ belongs to the proof; it is not the residual $F(x) + F'(x)s$ or its norm. Its increment depends on the allowed tolerance and the preceding residual bound. With $\theta = 0$ and $\epsilon_0 = 0$, the recurrence gives $\epsilon_n = 0$. This does not extend to an initial value $\epsilon_0 > 0$, nor does it mean that choosing an exact step under a positive tolerance forces the slack to vanish.

Its Theorem 5.1 establishes that ϵ_n is nondecreasing and bounded. For $\theta > 0$ and a positive initial residual bound, $\epsilon_1 > \epsilon_0 \geq 0$, so $\sum \epsilon_n$ does not converge. The increments can, however, be summable. The quantity tending to zero is $f(t_n) + \epsilon_n$, not necessarily ϵ_n separately. Here, instead, $\sum \widehat{\Delta}_n < \infty$ and $\sum \widehat{E}_n < \infty$ are proved. The difference describes the role of the variables; it does not by itself establish superiority between the theorems.

In its application to analytic functions, page 361, it uses

$$f(t) = \frac{t}{1 - \gamma t} - 2t + b = b - t + \frac{\gamma t^2}{1 - \gamma t},$$

the same rational form as our majorant. The assumptions and the methods controlled are different, so their criteria are not identified, but this rational form is not claimed as a new feature of the present construction.

S6.3 Rewriting as inexact Newton: scope and limits

For the total step $s_n^{\mathrm{tot}} = x_{n+1} - x_n = p_n + r_n$, we have $F'(x_n)p_n = -F(x_n)$ and $F'(x_n)r_n = -Q_n$. Hence

$$F(x_n) + F'(x_n)s_n^{\mathrm{tot}} = -Q_n. \tag{S6.2}$$

The exact execution of a multipoint method does not prevent its total step from being represented as an inexact correction relative to another algorithm, in this case one-step Newton. A uniform admissible relative tolerance does not follow from (S6.2). One would also have to check

$$\|F'(x_0)^{-1}Q_n\| \le \theta\|F'(x_0)^{-1}F(x_n)\|$$

at all indices, with a tolerance satisfying the assumptions of the result invoked. Two separate upper bounds for the numerator and denominator do not prove that relative comparison. The existence and location of y_n, z_n needed to construct Q_n also require their own argument. Neither the impossibility nor the general validity of that semilocal reduction is claimed here.

S6.4 Meaning and logical role of our defects

The hatted quantities are constructed from α through the polynomial first cycle and the rational recurrences in Section S3. The sequences t_n, s_n, v_n are obtained by applying the same scalar scheme to g. They are compared through

$$\widehat{\Delta}_n = \frac{[\widehat{\varepsilon}_n - \gamma g(t_n)]_+}{\gamma}, \qquad \widehat{E}_n = \frac{[\widehat{A}_n + \widehat{C}_n - \gamma(t_{n+1} - t_n)]_+}{\gamma}.$$

For these fixed controls, they are the smallest nonnegative slacks, since

$$\begin{aligned} \gamma(g(t_n) + \widehat{\Delta}_n) &= \max\{\gamma g(t_n), \widehat{\varepsilon}_n\}, \\ \gamma(t_{n+1} - t_n + \widehat{E}_n) &= \max\{\gamma(t_{n+1} - t_n), \widehat{A}_n + \widehat{C}_n\}. \end{aligned} \tag{S6.3}$$

The actual norms can be smaller than their bounds: a positive defect does not prove that the actual residual violates exact majorization. Nor is it a realized physical or computational perturbation.

Summability is not obtained merely by writing (S6.3). First, closure of the controls and $\sum_{n\ge 0}(\widehat{A}_n + \widehat{C}_n) = \widehat{\tau}_\infty < \infty$ are proved. The inequalities $\widehat{\varepsilon}_n \le \widehat{A}_n$, $\gamma\widehat{\Delta}_n \le \widehat{\varepsilon}_n$, and $\gamma\widehat{E}_n \le \widehat{A}_n + \widehat{C}_n$ then give the convergent series, also from $n = 0$.

The error bounds have different roles. Summing the second identity in (S6.3),

$$\begin{aligned} \gamma\left(t_* - t_n + \sum_{k=n}^{\infty}\widehat{E}_k\right) &= \widehat{\tau}_\infty - \widehat{\tau}_n \\ &\quad + \sum_{k=n}^{\infty}[\gamma(t_{k+1} - t_k) - (\widehat{A}_k + \widehat{C}_k)]_+. \end{aligned} \tag{S6.4}$$

The radius bound does not exceed the one written with defects. The latter retains the reference to the scalar trajectory; the former uses the constructed controls directly. Both depend only on the initial data and are a priori. The estimate using a verified residual $\|G(x_n)\|$ is a posteriori. The quantitative improvement is not attributed to the mere use of the positive part.

S6.5 Contribution supported by these comparisons

The previous works contain ingredients of a perturbed majorization theory. We do not claim that every additive term is new or that the literature lacks deterministic controls for exact algorithms. The comparison does not assert priority throughout the literature.

The construction combines a fixed scalar reference of the three-step method, residual bounds that retain the cancellations, stage radii, a specific estimate for the first cycle, and finite tail control. Its result is the criterion $\beta\gamma \leq 1547/10000$, the error bounds, and the quantitative comparison in Section S3. Applying a more general formalism might require precisely these estimates. Until such an application is proved, neither complete equivalence nor irreducibility is claimed.

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